

Stem Cells in Tissue Homeostasis and Regeneration

CHAPTER 22

Cells evolved originally as free-living individuals, and such cells still dominate Earth and its oceans. But the cells that matter most to us, as humans, are specialized members of a multicellular community. These cells have lost features needed for independent survival and acquired peculiarities that serve the needs of the body as a whole. Although they share the same genome, they are spectacularly diverse in structure, chemistry, and behavior. There are more than 200 different named cell types in the human body that collaborate with one another to form many different tissues, which are arranged into organs performing widely varied functions. To understand them, it is not enough to analyze cells in a culture dish: we need also to know how they live, work, and die in their natural habitat, the intact body.

In Chapters 7 and 21, we saw how the various cell types become different in the embryo and how cell memory and signals from their neighbors enable them to remain different thereafter. In Chapter 19, we discussed the mechanisms that cells use to self-assemble into multicellular tissues, including the use of molecular devices that bind cells together and the extracellular materials that give tissues and organs support. But the adult body is not static: it is a structure in dynamic equilibrium, where new cells are continually being born, differentiating, and dying. Homeostatic mechanisms maintain a proper balance, so that tissue architecture is preserved despite the constant replacement of old cells by new.

In this chapter, we focus on these homeostatic processes that continue throughout life. In doing so, we illustrate some of the diversity of differentiated cell types, examining in particular the part played in many adult tissues by **stem cells**—undifferentiated cells that are specialized to provide a fresh supply of differentiated cells where these need to be continually replaced or when they are required in large number for tissue repair and regeneration. We will see that many adult tissues constantly renew and repair themselves, but others do not, in which case, lost cells are lost forever, causing deafness, blindness, dementia, and other disorders. We discuss how stem cells are maintained within a self-renewing tissue and how the fate of their daughter cells is determined. We then describe the remarkable ability of some animal species to regenerate entire limbs or organs, in one extreme example, reproducing all of the tissues of the organism from a single stem cell. In the final section of the chapter, we discuss how stem cells can be generated and manipulated artificially, raising the practical question that underlies a current storm of interest in stem-cell technology: How can we use our understanding of the processes of cell differentiation and tissue renewal to improve upon nature and make good those injuries and degenerations associated with disease and aging of the human body that have hitherto seemed beyond repair?

STEM CELLS AND TISSUE HOMEOSTASIS

In self-renewing adult tissues, cells are continually being born, differentiating, and dying. This “flow” of cells can be compared with the flow of water in a river: the river may look the same from day to day, but, as it flows downstream, the water

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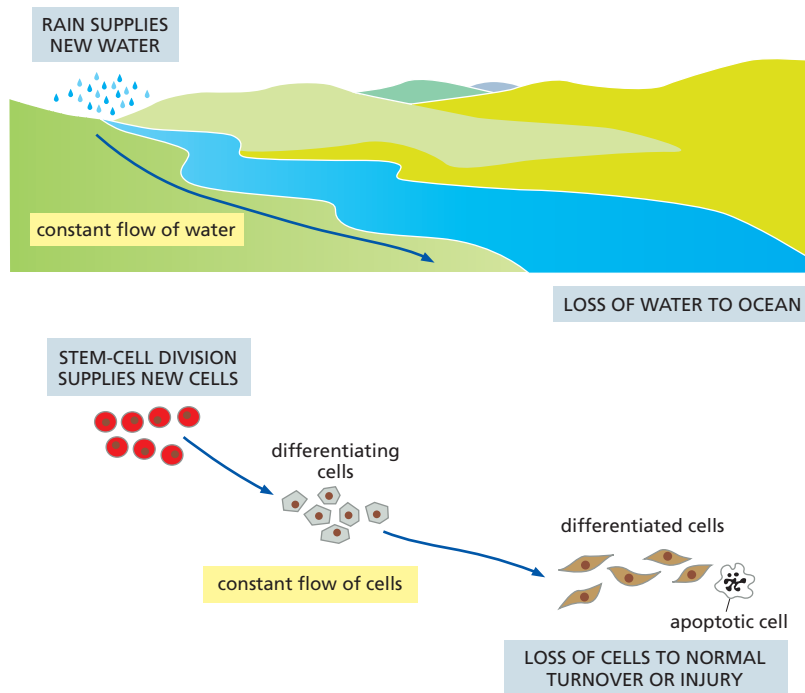


Figure 22-1 Tissue homeostasis compared to a river. In a self-renewing adult tissue, a constant flow of new cells produced by cell division “upstream” and a constant loss of differentiated cells “downstream” maintain the tissue in a dynamic equilibrium.

in the river is never the same (**Figure 22-1**). Similarly, the characteristic architecture of a self-renewing tissue is maintained even though the cell population is always changing, with differentiated cells lost “downstream” and stem cells producing new cells “upstream.” Thus, the stem cells in such tissues must be able to both replicate themselves and produce differentiated cells for the life span of the organism. Without stem cells, these tissues and organs would fail rapidly, being unable to keep up with the natural turnover of cells. Indeed, defects in stem-cell functions can contribute to disease and aging for this reason.

In this first section, we define the essential characteristics of stem cells, using two epithelial tissues as examples, the lining of the gut and the outer surface of the skin. We then consider how we can identify stem cells in tissues—by cell lineage tracing or, in the case of blood cells, by cell transplantation studies. We finally discuss tissues that can maintain themselves in the absence of stem cells.

Stem Cells Are Defined by Their Ability to Self-renew and Produce Differentiated Cells

Many adult tissues, especially those with high cell turnover rates, contain **tissue-specific stem cells** (also called **adult stem cells**). Each of these tissues contains its own unique stem-cell population, capable of producing the differentiated cells characteristic of that tissue and not those of other tissues. The stem cells of each tissue possess their own distinct developmental history, without necessarily sharing molecular characteristics with the stem cells of other tissues.

All stem cells, however, share two fundamental properties that define them: (1) they are able to replenish themselves as stem cells, generally throughout the lifetime of the organism—a process called *self-renewal*; and (2) they can also produce differentiated cells. Thus, when a stem cell divides, each daughter has a choice: it can either remain a stem cell or it can embark on a course that commits it to differentiation (**Figure 22-2**).

Stem cells usually do not produce differentiated cells directly; instead, they make an intermediate cell type that is committed to a differentiation pathway but continues to proliferate, thereby generating greater numbers of differentiated cells. These cells are called *progenitor cells*; they are also called *transit-amplifying cells* because their divisions serve to amplify the number of differentiated cells that

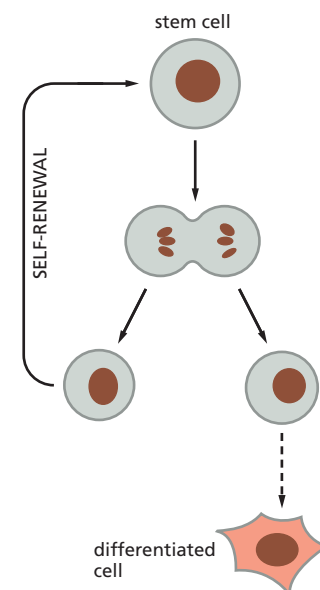


Figure 22-2 The defining characteristics of a stem cell. Each daughter cell produced when a stem cell divides can either remain a stem cell, in the process of self-renewal, or commit to differentiation, usually after a number of cell divisions. The self-renewal process maintains the pool of stem cells in the tissue.

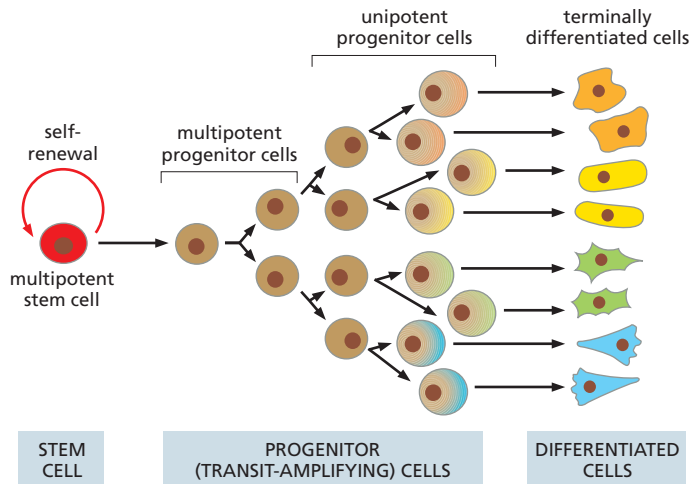


Figure 22-3 A hierarchy of stem cells, progenitor cells, and differentiated cells. In addition to self-renewal, tissue-specific stem cells generally produce progenitor (transit-amplifying) cells that divide a limited number of times before they terminally differentiate. Stem cells and progenitor cells can be unipotent or multipotent, depending on whether they produce only one type or multiple types of differentiated cells.

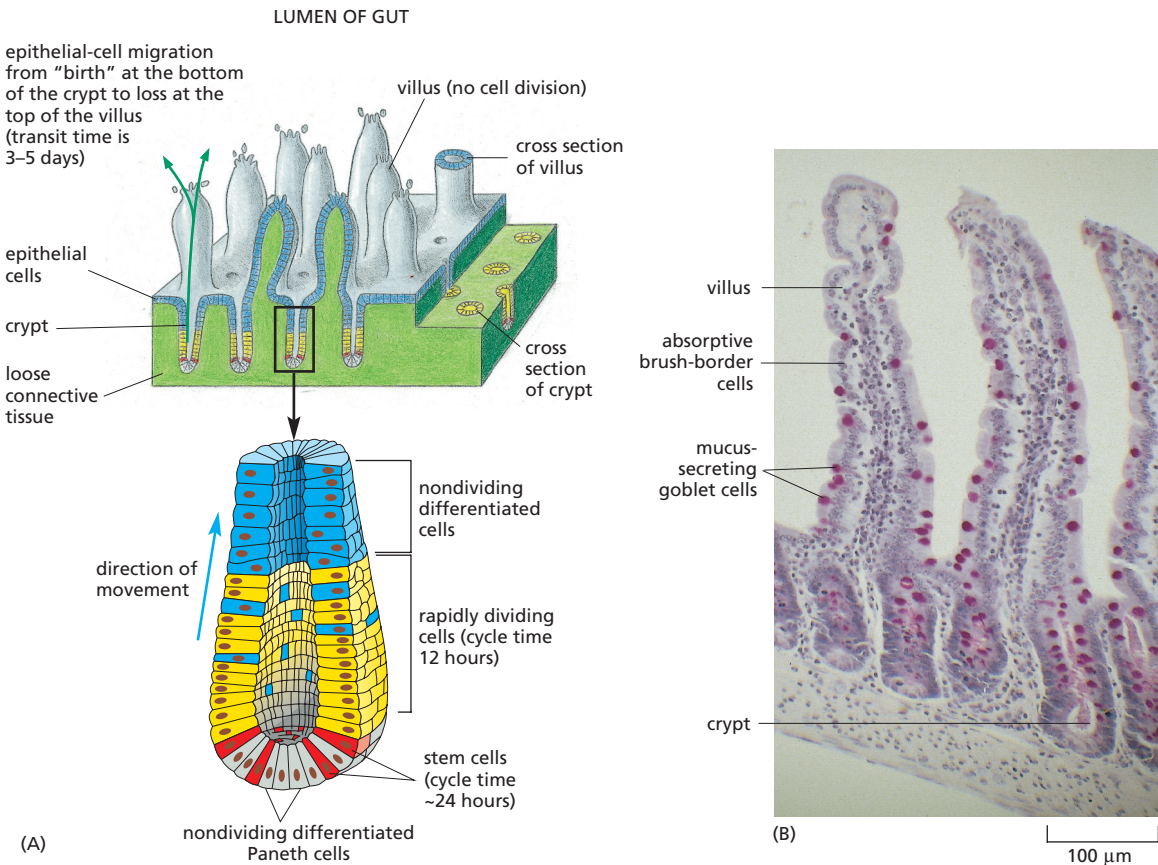
ultimately result from each stem-cell division (Figure 22-3). Unlike stem cells, these intermediate cells only go through a limited number of divisions before they differentiate. When a cell reaches the end of its differentiation pathway and does not divide again, it is *terminally differentiated*. When an adult stem cell or progenitor cell generates multiple differentiated cell types, it is *multipotent*; when it generates only one differentiated cell type, it is *unipotent* (see Figure 22-3).

We now consider two epithelial tissues to illustrate how these different categories of cells are organized in a self-renewing adult tissue—the lining of the gut and the outer layer of the skin. In both cases, the stem cells continually produce transit-amplifying progenitor cells at one surface of the tissue, while terminally differentiated cells are lost from the opposite surface—upstream and downstream, respectively, in our river analogy.

The Epithelial Lining of the Small Intestine Is Continually Renewed Through Cell Proliferation in Crypts

The lining of the small intestine (and of most other regions of the gut) is a single-layered epithelium, only one cell thick. The **intestinal epithelium** covers the surfaces of the *villi* that project into the gut lumen, and it lines the *crypts* that descend into the underlying connective tissue (Figure 22-4). Dividing cells, including the stem cells and progenitor cells, are restricted to the crypts, and terminally differentiated cells pour out of the crypts in a steady stream onto the villi. There are four main types of differentiated cells—one absorptive and three secretory (Figure 22-5):

1. **Absorptive cells** (also called *brush-border cells* or *enterocytes*) are the majority cell type in the epithelium and have densely packed microvilli on their exposed surfaces. Their job is to take up nutrients from the gut lumen. To this end, they also produce hydrolytic enzymes that perform some of the final steps of extracellular digestion of food.
2. **Goblet cells** secrete mucus into the gut lumen; this mucus covers the epithelium with a protective coat.
3. **Paneth cells** form part of the innate immune defense system (discussed in Chapter 24) and secrete proteins that kill bacteria; they also secrete Wnt signal proteins (discussed in Chapter 15) required to maintain the stem-cell population.
4. **Enteroendocrine cells**, of more than 15 different subtypes, secrete serotonin and peptide hormones that act on neurons and other cell types in the gut wall and regulate the growth, proliferation, and digestive activities of cells of the gut and other tissues.



As if on a conveyor belt, the absorptive, goblet, and enteroendocrine cells travel upward from their site of birth in the crypt, by a sliding movement in the plane of the epithelial sheet, to cover the surfaces of the villi. Within 3–4 days (in the mouse) after emerging from the crypts, the differentiated cells reach the tips of the villi, where they are discarded into the gut lumen (see Movie 20.7). The Paneth cells in the crypts are produced in much smaller numbers and have a different migration pattern. They live at the bottom of the crypts, where they too are continually replaced, although not so rapidly, persisting for several weeks before undergoing apoptosis and being phagocytosed by their neighbors.

The stem cells that give rise to the intestinal epithelium are located just above the base of the crypt, interspersed among the Paneth cells. The stem cells can be identified because only they express a particular G-protein-coupled receptor called *Lgr5*, which serves as a specific marker for the stem cell population. When the stem cells divide, which they do every 24 hours or so in the mouse intestine, some of the progeny commit to differentiation, becoming transit-amplifying progenitor cells that migrate upward, while others remain stem cells in the process of self-renewal. The stem cells are multipotent, producing all the differentiated cell types in the epithelium.

Epidermal Stem Cells Maintain a Self-renewing, Waterproof, Epithelial Barrier on the Body Surface

Stem-cell systems are organized in a variety of different ways depending on the tissue. For example, the outer epithelial covering of the body, the **epidermis**, undergoes continual renewal, but, unlike the intestinal epithelium, it is multi-layered, or *stratified*. Stem cells are located in the basal layer, as are the dividing transit-amplifying progenitor cells. Once the progenitor cells stop dividing, they leave the basal layer and move outward toward the exposed surface, undergoing

Figure 22-4 Renewal of the gut epithelial lining. (A) The pattern of cell turnover and proliferation in the epithelium that forms the lining of the small intestine. Stem cells (red) lie at the crypt base, interspersed among nondividing differentiated cells (Paneth cells). Progeny of the stem cells move mainly upward from the crypts onto the villi; after a few quick divisions, they cease dividing and differentiate—some of them while still in the crypt, most of them as they emerge from the crypt. The Paneth cells, like the other nondividing differentiated cells, are continually replaced by the progeny of the stem cells, but they migrate downward to the crypt base and survive there for many weeks. (B) Micrograph of a section of part of the lining of the small intestine, showing the crypts and villi. Note the mixture of differentiated cell types, all generated from the stem cells; these are primarily absorptive cells, with mucus-secreting goblet cells (stained red) interspersed among them.

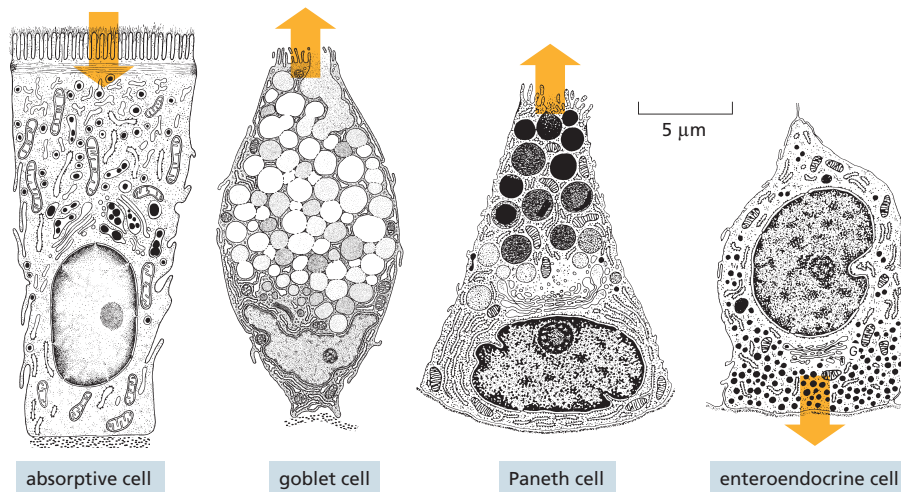


Figure 22-5 The four main differentiated cell types found in the epithelial lining of the small intestine. All cells are oriented with the gut lumen at *top*. Broad orange arrows indicate direction of secretion or uptake of materials for each type of cell. All of these cells are generated from undifferentiated multipotent stem cells living near the bottoms of the crypts (see Figure 22-4). *Absorptive (brush-border) cells* outnumber the other cell types in the epithelium by about 10:1 or more. The microvilli on their apical surface provide a 30-fold increase in surface area, not only for the import of nutrients but also for the anchorage of enzymes that perform the final stages of extracellular digestion, breaking down small peptides and disaccharides into monomers that can be transported across the cell membrane. *Goblet cells* secrete mucus; these are the most common of the secretory cell types. *Paneth cells* secrete (along with some growth factors) *cryptdins*—proteins of the defensin family that kill bacteria. Different subtypes of *enteroendocrine cells* secrete serotonin and peptide hormones into the gut wall (and thence the blood). Cholecystokinin is a hormone released from enteroendocrine cells in response to the presence of nutrients in the gut. It binds to receptors on nearby sensory nerve endings and causes the release of digestive enzymes from the pancreas and bile from the gall bladder; it also signals to the brain to stop the feeling of hunger once one has eaten enough. (Absorptive and goblet cells, Don W. Fawcett/Science Source; Paneth and enteroendocrine cells, from R.V. Krstić, *Illustrated Encyclopedia of Human Histology*. Berlin: Springer-Verlag, 1964. With permission from Springer Nature.)

terminal differentiation as they go. They end up as lifeless scales, or *squames*, which are eventually shed from the surface of the skin (**Figure 22-6**).

Even though the architecture of the epidermis is very different from that of the intestinal epithelium, many of the same basic principles apply. The stem cells are maintained by signals within a specific region of the tissue, which in the epidermis is the basal lamina and underlying connective tissue. The daughters of stem cells that are committed to differentiation undergo several divisions as transit-amplifying

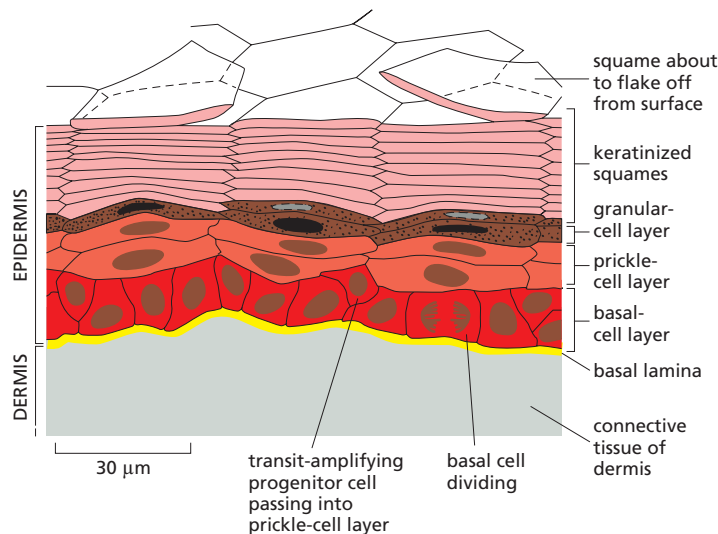


Figure 22-6 The multilayered structure of the epidermis. The epidermis forms the outer covering of the skin, creating a waterproof barrier that is self-repairing and continually renewed. Beneath this lies a relatively thick layer of connective tissue, which includes the tough, collagen-rich dermis (from which leather is made). The cells of the epidermis are called keratinocytes, because their characteristic differentiated activity is the synthesis of keratin intermediate filament proteins, which give the epidermis its toughness. These cells change their appearance and properties from one layer to the next, progressing through a regular program of terminal differentiation. Those in the innermost layer, attached to an underlying basal lamina, are termed basal cells, and it is normally only these that divide: the basal-cell population includes relatively small numbers of stem cells along with larger numbers of transit-amplifying progenitor cells derived from them. Above the basal cells are several layers of larger prickle cells. Beyond the prickle cells lies the thin, darkly staining granular-cell layer, where the cells are sealed together to form a waterproof barrier; this marks the boundary between the inner, metabolically active strata and the outermost layer of the epidermis, consisting of dead cells whose intracellular organelles have disappeared. These outermost cells are reduced to flattened scales, or squames, filled with densely packed keratin, which are eventually shed from the surface of the skin. The time from exit of a cell from the basal layer to its loss by shedding at the surface is a week or two, depending on body region and species.

cells in the basal layer before differentiating. Moreover, most of the signaling pathways that organize the intestinal stem-cell system are also involved in regulating the epidermal stem-cell system, although with different individual roles.

Cell Lineage Tracing Reveals the Location of Stem Cells and Their Progeny

Stem cells in adult tissues are usually rare and difficult to identify in conventional tissue sections, unless a stem cell-specific marker like *Lgr5* is available. Recombinant DNA technology provides a general and powerful way to identify stem cells and their progeny in any renewing tissues using a technique called *cell lineage tracing*. The method uses transgenic animals to create a visible genetic mark in just a few cells, which, over time, give rise to widely separated and easily distinguished clones of progeny cells, as explained in [Figure 22-7](#) and [Figure 22-8](#). This approach does not require prior knowledge as to whether a tissue contains stem cells or not. If they exist, the stem cells will be marked randomly and will lead to a persistent clonal lineage that contains stem cells as well as differentiated cells (see [Figure 22-7B](#)); dividing progenitor cells will also be randomly marked and produce labeled clones, but all of these will eventually disappear (see [Figure 22-7C](#)). The analysis of the clones not only indicates whether stem cells are

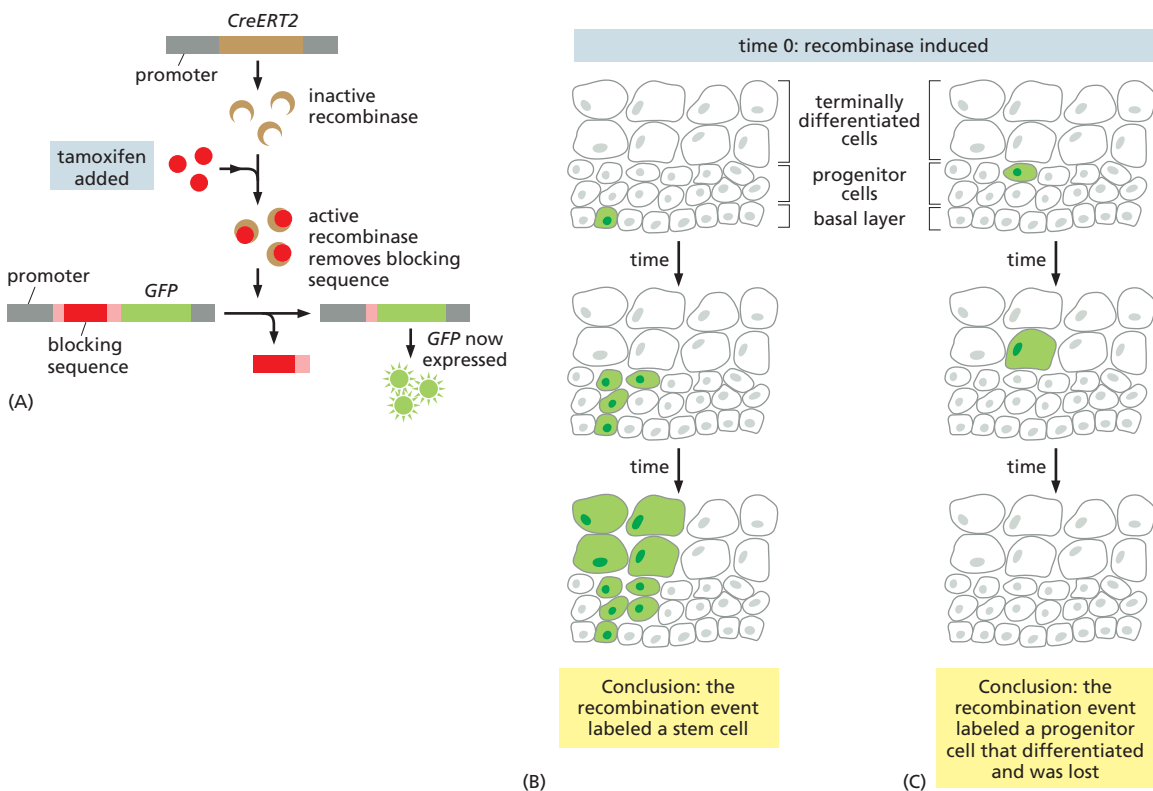


Figure 22-7 Clonal analysis using a genetic marker. (A) Transgenic animals containing two transgenes can be used to drive expression of a readily detected and heritable marker in a small set of cells. The first transgene encodes a marker gene, such as one that encodes green fluorescent protein (GFP). However, the expression of the *GFP* transgene, shown here in green, is prevented by a blocking sequence (red) that is flanked by LoxP sites (pink; see [Figure 5-66](#)). The second transgene, *CreERT2* (brown), encodes a chimeric form of the Cre recombinase called CreERT, which consists of the Cre recombinase linked to the estrogen receptor protein; this enzyme becomes active as a recombinase only when it binds the artificial estrogen analog tamoxifen (red spheres). Addition of tamoxifen leads to a recombination event that removes the blocking DNA sequence. As a result, the *GFP* marker is expressed. Because the blocking DNA has been permanently removed from the genome, the marker continues to be expressed in all the descendants of a cell in which the recombination event has occurred. With a low dose of the inducer molecule tamoxifen, it is possible to activate the marker at random in just a few widely spaced cells, giving rise to distinguishable clones. (B) If the recombination event occurs in a stem cell, a clonal lineage will be marked, and the labeling will persist over time as the marked stem cell self-renews and produces differentiated cells. (C) If the recombination event occurs in a cell that is not a stem cell, the label will disappear over time as the marked cell differentiates and is eventually lost.

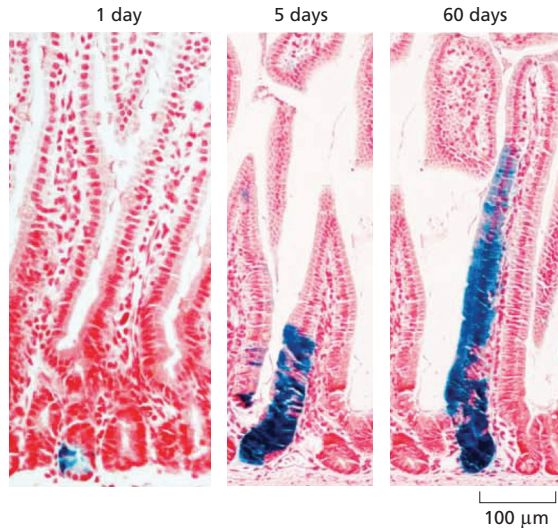


Figure 22-8 Lgr5-expressing stem cells and their progeny in the small intestine. The basic method shown in Figure 22-7 was modified here to mark single intestinal stem cells and trace the fates of their progeny. The *Lgr5* gene encodes a member of the family of G-protein-linked transmembrane receptors, and it is expressed specifically in stem cells near the crypt base. In this case, the *Lgr5* promoter was used to drive expression of *CreERT2*, and treatment with a low dose of tamoxifen resulted in occasional stem cells expressing the marker protein LacZ (rather than GFP). These cells and all of their progeny could subsequently be detected with a *blue* histochemical stain. All of the *blue* cells in these images derive from a single *Lgr5*-expressing stem cell. After 60 days, the *blue* progeny of this cell are seen to extend almost all the way up a villus. These progeny can be shown to include all types of differentiated cells, as well as persistent *Lgr5*-expressing cells at the crypt base. This proves that *Lgr5*-expressing cells are multipotent stem cells. (From N. Barker et al., *Nature* 449:1003–1007, published 2007 by Nature Publishing Group. Reproduced with permission of SNCSC.)

present, but also where the stem cells are located and whether they are unipotent or multipotent.

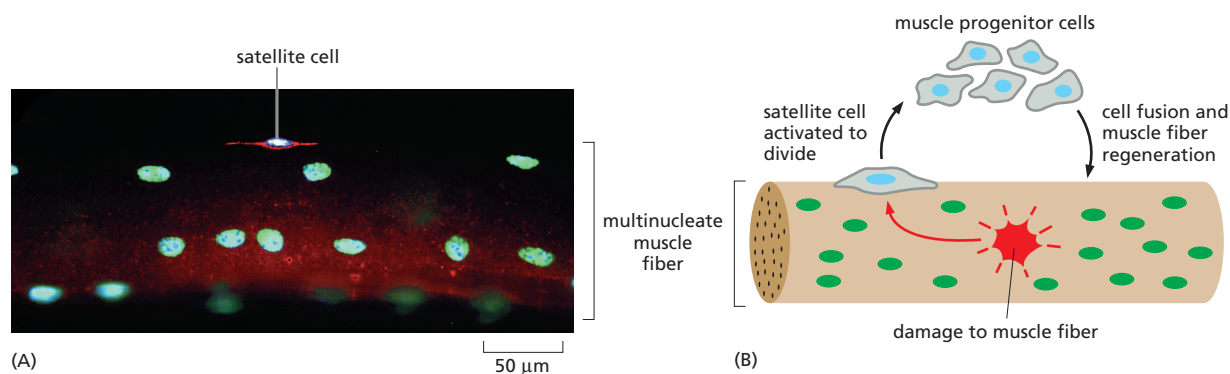
A more directed approach can be used for cell lineage tracing if a gene that is expressed specifically in the stem cells in a tissue is known, as is the case for the *Lgr5* gene in the mouse intestine. In this case, one can use the gene's promoter to express the genetic marker specifically in the stem cells. It was this type of experiment that initially established that *Lgr5*-expressing cells in the mouse intestinal epithelium are stem cells and that they are multipotent (see Figure 22-8).

Quiescent Stem Cells Are Difficult to Identify by Lineage Tracing

The lineage tracing method just described assumes that the stem cells in a tissue are actively dividing to generate daughter cells that self-renew or differentiate. Some adult stem cells, however, reside in a quiescent state, serving as a “reservoir” for when they are needed: they divide only rarely or not at all, unless they are induced to do so by a stimulus, such as tissue injury. In these cases, it can require extra time or stimulation to reveal the stem cells by lineage tracing.

Human **skeletal muscle** provides an example. It consists of multinucleated muscle cells (muscle fibers) that form during development by the fusion of terminally differentiated *myoblasts*. Humans do not normally generate new skeletal muscle in adult life, but they still have the capacity to do so when there is a need for muscle growth or repair. Cells capable of serving as myoblasts are retained as small, flattened, and nondividing cells lying in close contact with the mature muscle fiber and contained within its sheath of basal lamina (**Figure 22-9**). If the muscle is damaged or stimulated to grow, these *satellite cells* are activated to proliferate, and their progeny can fuse with the existing muscle fiber to repair

Figure 22-9 The repair of skeletal muscle fibers by satellite cells. (A) The specimen is stained with an antibody (*red*) against a muscle cadherin, M-cadherin, which is present on both the satellite cell and the muscle fiber and is concentrated at the site where their membranes are in contact. The nuclei of the muscle fiber are stained *green*, and the nucleus of the satellite cell is stained *blue*. (B) Schematic drawing of the repair of a damaged muscle fiber by the proliferation and fusion of satellite cells. (A, courtesy of Terence Partridge.)



the damaged muscle or to allow muscle growth. Satellite cells or some subset of them are thus the stem cells of adult skeletal muscle, normally held in reserve in a quiescent state but available when needed as a self-renewing source of terminally differentiated myoblasts.

The process of muscle repair by means of satellite cells is limited in what it can achieve. In one form of *muscular dystrophy*, for example, a genetic defect in the cytoskeletal protein dystrophin slowly but progressively damages differentiated skeletal muscle cells. As a result, satellite cells proliferate to repair the damaged muscle fibers. But this regenerative response is unable to keep pace with the damage, and connective tissue eventually replaces the muscle fibers, blocking any further possibility of repair. A similar decline in the capacity for repair contributes to the progressive muscle weakening that occurs in the elderly.

Hematopoietic Stem Cells Can Be Identified by Transplantation

A different method to identify stem cells in adult tissues is by cell transplantation. The method was first used to identify the stem cells of the *hematopoietic* (blood-making) system, the most complex stem-cell system in the adult mammalian body. The **hematopoietic stem cells** that give rise to both the red blood cells (**erythrocytes**) and white blood cells (**leukocytes**) are located in the adult bone marrow, where they also produce blood *platelets*. There are many different white blood cell types, including monocytes that can exit the bloodstream and develop into *macrophages*, which are found in most organs. When an animal is exposed to a large dose of x-rays, most of the hematopoietic cells in the bone marrow are destroyed, and, as a result, the animal dies within days because of its inability to produce new blood cells. The animal can be saved, however, by a transfusion of cells taken from the bone marrow of a healthy donor mouse of the same inbred

TABLE 22–1 Blood Cells

Type of cell	Main functions	Typical concentration in human blood (cells/liter)
Red blood cells (erythrocytes)	Transport O ₂ to and CO ₂ from tissues	5×10^{12}
White blood cells (leukocytes)		
<i>Granulocytes</i>		
Neutrophils (polymorphonuclear leukocytes)	Phagocytose and kill invading bacteria	5×10^9
Eosinophils	Destroy larger parasites and modulate allergic inflammatory responses	2×10^8
Basophils	Release histamine (and in some species serotonin) in certain immune reactions	4×10^7
<i>Monocytes</i>	Become tissue macrophages, which phagocytose and digest invading microorganisms and foreign bodies as well as damaged senescent cells; some also differentiate into dendritic cells	4×10^8
<i>Lymphocytes</i>		
B cells	Make and secrete antibodies	$\sim 0.3 \times 10^9$
T cells	Kill virus-infected cells and regulate activities of other leukocytes	$\sim 2 \times 10^9$
Natural killer (NK) cells	Kill virus-infected cells and some tumor cells	1×10^8
Platelets (cell fragments arising from megakaryocytes in bone marrow)	Initiate blood clotting	3×10^{11}
Humans contain about 5 liters of blood, accounting for 7% of body weight. Red blood cells constitute about 45% of this volume and white blood cells about 1%, the rest being the liquid blood plasma.		

Figure 22–10 Rescue of an irradiated mouse by a transfusion of bone marrow cells. An essentially similar procedure is used in the treatment of leukemia in humans by bone marrow transplantation after irradiation or chemotherapy.

strain. Among these cells there are some that can colonize the irradiated host and permanently re-equip it with hematopoietic tissue (**Figure 22–10**). Such experiments prove that the bone marrow contains an entire hematopoietic stem-cell system, and they have allowed scientists to isolate the relevant stem cells and discover the molecular features that distinguish them.

For this purpose, cells taken from mouse bone marrow are sorted (using a fluorescence-activated cell sorter) according to their cell-surface antigens, and the different fractions are transfused into irradiated mice. If a fraction rescues the irradiated mice, it must contain hematopoietic stem cells. In this way, it was shown that the hematopoietic stem cells display a specific combination of cell-surface proteins and that, by appropriate cell sorting, one can obtain virtually pure stem-cell preparations. The stem cells turn out to be a tiny fraction of the mouse bone marrow population—about 1 cell in 50,000–100,000; but this is enough. Remarkably, a single such cell injected into a host mouse with defective hematopoiesis is sufficient to reconstitute its entire hematopoietic system, generating a complete set of blood-cell types, as well as fresh stem cells. This and lineage tracing experiments have established that an individual hematopoietic stem cell is *multipotent* and can self-renew and give rise to the complete range of blood-cell types.

Blood contains large numbers of many types of differentiated cells (**Table 22–1**), many of which can be seen in a standard, stained smear of human blood (**Figure 22–11**). Erythrocytes are homogeneous and remain in the blood vessels, where they transport O_2 and CO_2 bound to hemoglobin. By contrast, leukocytes are heterogeneous in morphology and function and must crawl across the walls of small blood vessels into tissues to function. Terminally differentiated

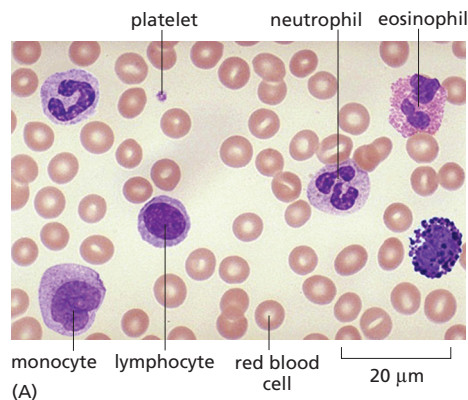
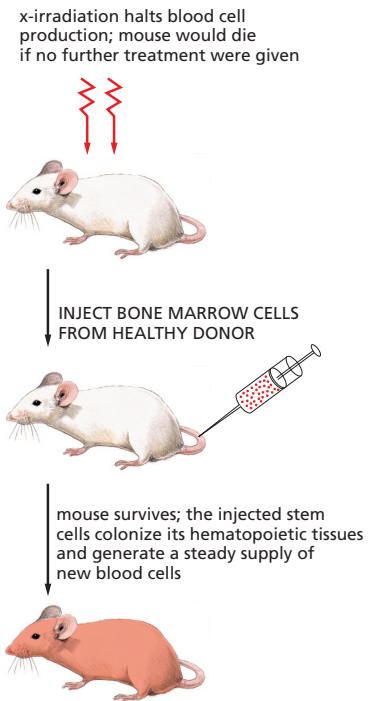
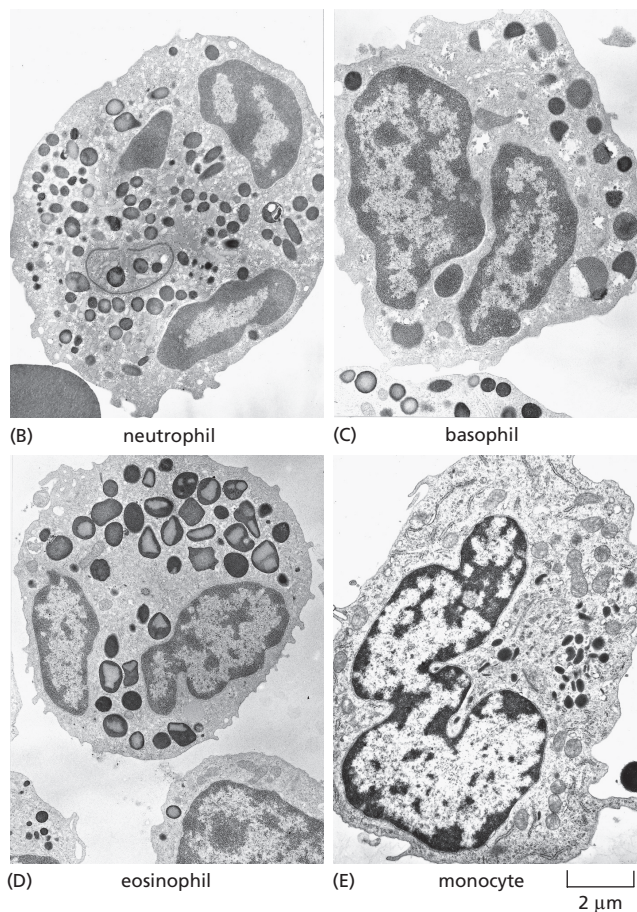


Figure 22–11 Human blood cells.

(A) A light micrograph of a blood smear stained with the Romanowsky stain, which faintly colors the red blood cells red and strongly colors the white blood cells blue. (B–E) Electron micrographs of (B) a neutrophil, (C) a basophil, (D) an eosinophil, and (E) a monocyte. (Electron micrographs of lymphocytes are shown in Figure 24–14.) Each of the cell types shown here has a different function (see Table 22–1), which is reflected in the distinctive types of secretory granules and lysosomes each cell type contains. There is only one nucleus per cell, but it has an irregular lobed shape, and in panels B, C, and D the connections between the lobes are out of the plane of section. (A–D, courtesy of Dorothy Bainton; E, courtesy of David Mason.)

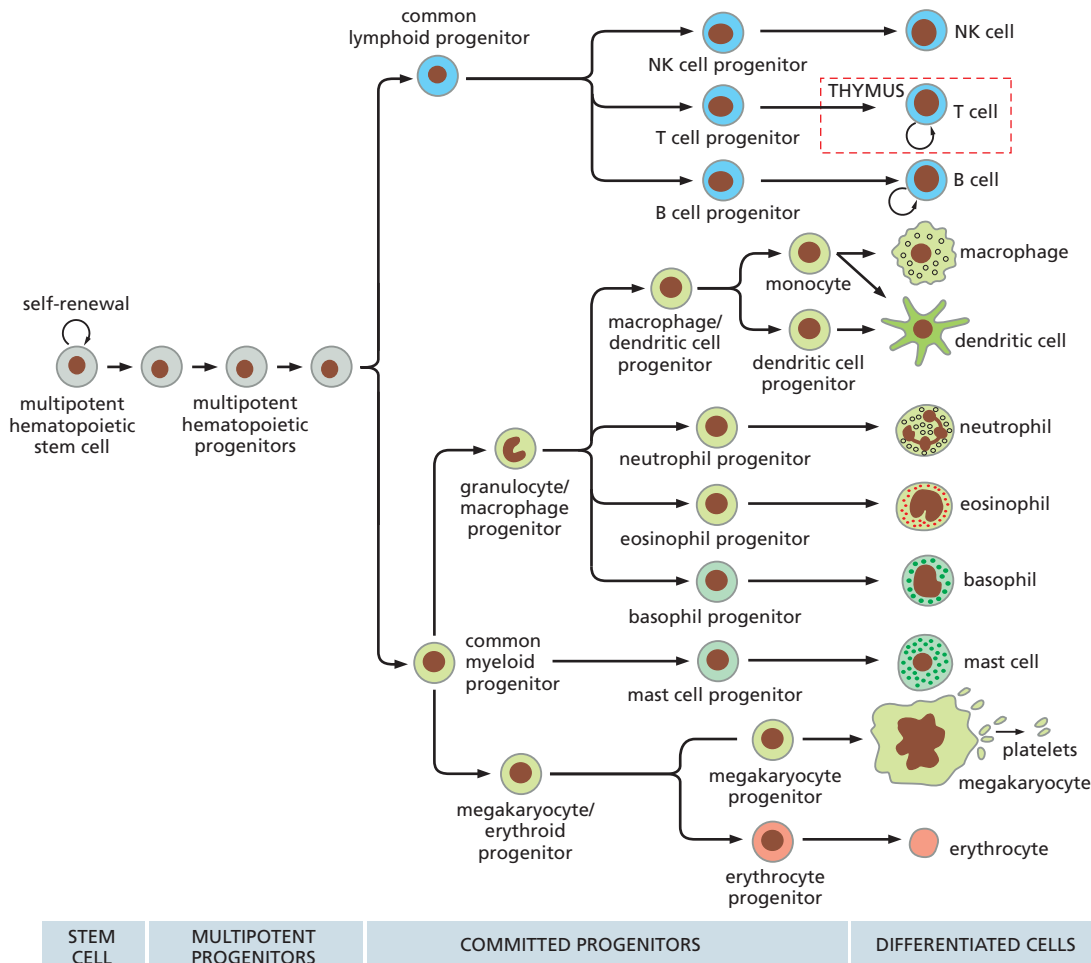


blood cells have relatively short life spans and are produced throughout the life of the animal, so hematopoietic stem cells must generate enormous numbers of these differentiated cells each day. But they do not produce them directly. Instead, the stem cells continually produce large numbers of transit-amplifying progenitor cells in the bone marrow, which are committed to differentiation but go through multiple divisions before they terminally differentiate.

The stem cells, however, do not jump directly from a multipotent stem-cell state into a committed and specific pathway of differentiation; instead, they go through a number of cell divisions, in which they progressively restrict their developmental options in a series of steps (Figure 22-12). The first step is usually to become committed to either a myeloid or a lymphoid fate, by way of two kinds of multipotent transit-amplifying progenitor cells. One is capable of generating large numbers of all the different types of **myeloid cells**, including blood **granulocytes** (neutrophils, eosinophils, and basophils), **monocytes** (the precursors of macrophages and dendritic cells), and **megakaryocytes** (which remain in the bone marrow and produce blood platelets by pinching off cell fragments) (Movie 22.1). The other type of multipotent progenitor cell gives rise to large numbers of different types of **lymphoid cells**, including the B and T lymphocytes of the adaptive immune system and the lymphocyte-like *natural killer* (NK) cells of the innate immune system (discussed in Chapter 24). Further commitment steps ultimately give rise to progenitor cells committed to the production of just one cell type, although this final commitment step occurs well before the cells cease proliferating and terminally differentiate. Many different signal molecules, produced inside and outside the bone marrow, control the survival, proliferation, and pathway commitment of hematopoietic cells and their committed progeny in the bone marrow and thereby regulate how many of each type of differentiated blood cell is eventually produced.

Figure 22-12 A simplified scheme of mouse and human hematopoiesis.

A multipotent hematopoietic stem cell normally divides infrequently to generate either more multipotent stem cells, which are self-renewing, or multipotent progenitor cells, which give rise to all the cells of the blood and immune system. The progenitor cells divide a limited number of times and go through multiple stepwise intermediates before they develop into fully differentiated cells. As they go through their divisions, the progenitors become progressively more specialized in the range of cell types that they can give rise to, as indicated by the branching of this cell lineage diagram. In adult mammals, all of the cells shown develop mainly in the bone marrow—except for T lymphocytes, which as indicated develop in the thymus, and macrophages and some dendritic cells, which develop from monocytes that are circulating in the blood. Note that not all stem cells generate the identical patterns of progeny via precisely the same sequence.



In this hierarchical hematopoietic system, only the stem cells can self-renew for the life of the individual, and a single stem-cell division can lead to the production of thousands to millions of differentiated progeny. This explains why the number of stem cells is such a tiny fraction of the total population of hematopoietic cells in the bone marrow. Keeping the number of stem-cell divisions low has important advantages. If these divisions on their own had to keep up with the high demand for terminally differentiated blood cells, they would result in rapid replicative cell senescence and exhaustion of the stem-cell pool, with dire consequences. Moreover, the lower the number of stem-cell divisions, the lower the risk of the cells accumulating dangerous mutations, which would persist in mutant clones; as discussed in Chapter 20, such clones are a particular danger in the hematopoietic system, where a relatively small number of mutations can be sufficient to cause blood-cell cancers.

Some Tissues Do Not Require Stem Cells for Their Maintenance

Some types of cells can divide even though fully differentiated, allowing for renewal and regeneration without the use of stem cells. The insulin-secreting **pancreatic β cells** are one example. Their mode of renewal has a special importance, because it is their loss through autoimmune attack that is responsible for type 1 (juvenile-onset) diabetes, and their functional decline with age and obesity is also a significant factor in type 2 (adult-onset) diabetes. The β cells are normally sequestered in cell clusters called *islets of Langerhans*. The islets seem not to contain stem cells, yet new β cells are continually generated within the islets. Lineage tracing studies, similar to those described earlier, show that the renewal of this population normally occurs by the simple division of differentiated insulin-producing β cells.

Another tissue that can renew by simple division of fully differentiated cells is the liver. The main cell type in the liver is the **hepatocyte**, a large cell that performs the liver's many metabolic functions. Hepatocytes normally live for a year or more and divide at a very slow rate. Powerful homeostatic mechanisms operate to adjust both their rate of cell proliferation and their rate of cell death, to keep the liver at its normal size and to restore that size in the case of damage. A dramatic effect is seen if large numbers of hepatocytes are removed surgically or killed by poisoning with carbon tetrachloride. Within a day or so after either sort of damage, a surge of cell division occurs among the surviving hepatocytes, quickly replacing the lost tissue. If two-thirds of a rat's liver is removed, for example, a liver of nearly normal size can regenerate by hepatocyte proliferation within about 2 weeks.

Both the pancreas and the liver also contain small populations of stem cells that can be called into play as a backup mechanism to produce the differentiated cell types in more extreme circumstances.

In Response to Injury, Some Differentiated Cells Can Revert to Progenitor Cells and Some Progenitor Cells Can Revert to Stem Cells

Although the pathways from stem cell to progenitor cell to differentiated cell are normally unidirectional, there are some cases where injury can reverse the direction. One striking example occurs when a myelinated mammalian nerve is cut: the axon distal to the cut degenerates, and the differentiated myelinating Schwann cells (see Figure 11–35) *dedifferentiate* to form proliferating Schwann-cell progenitor cells. These progenitor cells help guide the regenerating axons back to their original targets and then remyelinate the axons to complete the regeneration process.

Similarly, in some tissues, when stem cells are lost, progenitor cells that have committed to differentiation can reprogram to revert to stem cells. In both the mouse and *Drosophila* testis, for example, spermatogonial cells typically follow a unidirectional development pathway from stem cell to proliferating progenitor cells, which undergo meiosis and finally differentiate into sperm. If stem cells are lost, either naturally or experimentally, the mitotically proliferating progenitor

cells can reprogram and revert to stem cells. More generally, this process might contribute to the long-term maintenance of stem-cell populations, allowing the lifetime of individual stem cells to be shorter than the lifetime of the organism. Unfortunately, cancer cells often also acquire stem cell–like properties, allowing them to self-renew indefinitely (discussed in Chapter 20).

Some Tissues Lack Stem Cells and Are Not Renewable

Some adult tissues that lack stem cells are not able to regenerate. The remarkable variation in the ability of different tissues to regenerate is illustrated by comparing the olfactory epithelium in the nose, the auditory epithelium of the inner ear, and the photoreceptive epithelium of the retina, which exhibit striking differences in their renewal capacity. The *olfactory epithelium* contains a population of stem cells that give rise to differentiated cells that have a limited life span and are continually replaced. But unlike the epidermis discussed earlier, these differentiated olfactory cells are neurons; they have their cell bodies in the olfactory epithelium and extend their axons back to the olfactory bulbs in the brain. The renewal of this epithelium therefore involves the continual production of new axons that have to navigate back to specific sites in the brain, where they form new synapses.

In contrast, in mammals at least, the *auditory epithelium* and *retinal photoreceptive epithelium* lack stem cells, and their sensory receptor cells—the sensory hair cells in the ear and the photoreceptors in the retina—are irreplaceable. If they are destroyed—whether by too much exposure to loud noise, by looking into the beam of a laser, or through degenerative processes occurring in disease or in old age—the loss is permanent.

We will return to tissue regeneration later in the chapter.

Summary

Many adult tissues, particularly those with a high cell turnover rate such as the intestinal lining, skin epidermis, and blood, are continually renewed by stem cells to maintain tissue homeostasis throughout the lifetime of the organism. Stem cells are defined by their ability to both self-renew and to generate terminally differentiated cells, usually by way of rapidly dividing, transit-amplifying progenitor cells. These properties can be revealed experimentally through lineage-tracing or transplantation experiments. Tissue-specific, or adult, stem cells are restricted in their differentiation potential, only generating one or more of the specific cell types of a particular tissue.

In the single-layer lining of the small intestine, multipotent stem cells are located near the base of each crypt, where they self-renew and produce dividing committed progenitor cells, most of which flow upward and terminally differentiate into one of three main types of gut cells when they reach the villus; other progenitor cells move in the opposite direction and become Paneth cells, which remain at the base of the crypt and help maintain the stem cells. Other self-renewing epithelia, such as the epidermis, have a multilayered (stratified) architecture, with stem cells and their differentiating progeny arranged in different ways, but are governed by similar basic principles.

The hematopoietic system is the most complex mammalian stem-cell system; all the red blood cells and the many types of white blood cells derive from a common, multipotent, hematopoietic stem cell in the adult bone marrow, where it divides slowly and produces multipotent and unipotent progenitor (transit-amplifying) cells, which divide rapidly and differentiate into a large number and variety of terminally differentiated cell types every day. In other tissues, such as skeletal muscle, stem cells are quiescent and only divide and differentiate when tissue growth or repair is required. Adult tissue renewal and repair do not always depend on stem cells; in the pancreas and liver, for example, differentiated cells can divide throughout life to replace lost cells and maintain tissue homeostasis. In some cases, progenitor cells can be generated from differentiated cells, and stem cells can be

generated from progenitor cells. At the opposite extreme, some sensory epithelia of the adult mammalian ear and eye are not renewable; their sensory cells do not undergo turnover—once lost, they are lost forever.

CONTROL OF STEM-CELL FATE AND SELF-RENEWAL

Adult stem cells maintain tissue homeostasis in renewing tissues, owing to their ability both to self-renew and to produce differentiated cells for the lifetime of the organism. How do these cells maintain their stem-cell identity for so long, and how do their daughter cells choose between self-renewal and a commitment to differentiation? While maintaining a stem-cell population is critically important for self-renewing tissues, stem-cell proliferation must be kept in check, as uncontrolled production of undifferentiated cells is a hallmark of cancer (discussed in Chapter 20). In this section, we discuss the mechanisms that allow stem cells to maintain their identity while producing differentiated cells and keeping their own proliferation under control.

The Stem-Cell Niche Maintains Stem-Cell Self-Renewal

The identity and self-renewal of stem cells depend on extracellular signals from their environment that both promote their proliferation and inhibit their commitment to differentiation. In many tissues, stem cells inhabit a special microenvironment called the **stem-cell niche**, where these signal molecules are provided in high concentration (Figure 22-13). The stem cells remain in close physical proximity to the niche-supporting cells, which produce the signals and provide a local special environment. Outside this environment, the concentration of the signal molecules is insufficient to maintain the self-renewal of the stem cells, and their daughter cells therefore commit to differentiation.

The existence of the stem-cell niche was first inferred from the experiment in which bone marrow cells were transplanted into a mouse that had been subjected to a high dose of x-rays (see Figure 22-10). Without the prior irradiation, the transplanted cells were unable to reestablish hematopoiesis, suggesting that the resident hematopoietic stem cells first had to be removed before the transplanted stem cells could successfully seed the host's bone marrow. The hypothetical special "place" occupied by either the host or donor hematopoietic stem cells was referred to as the *stem-cell niche*. Because of the complexity of bone marrow and the rarity of the stem cells there, the location of the niche remained unclear until many years later, when immunofluorescence microscopy and the use of cell markers provided evidence that the surface of sinusoids in the bone marrow is the likely site of the niche.

The molecular nature of stem cell-renewal signals provided within the niche varies across species and tissues, but they frequently are secreted signal proteins of the Wntless (Wnt), Hedgehog (Hh), or transforming growth factor (TGF) families (discussed in Chapter 15). Other niche signals are cell-surface proteins that depend on direct contact between the stem cells and the niche-supporting cells. The mammalian intestinal crypt provides a well-studied example (see

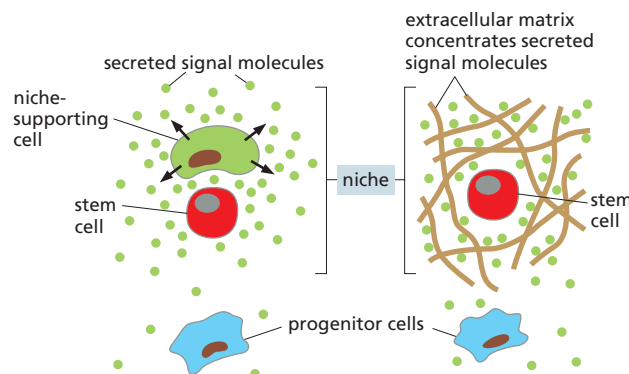


Figure 22-13 The stem-cell niche.

Stem cells undergo self-renewal only in a specialized microenvironment where they are exposed to the necessary signal molecules. These signal molecules can be provided by niche-supporting cells or by a specialized extracellular matrix that serves to concentrate them. The niche environment can be very small. In some cases, a stem cell directly adjacent to the niche is capable of self-renewal, while a cell just one cell-diameter away from the niche is not.

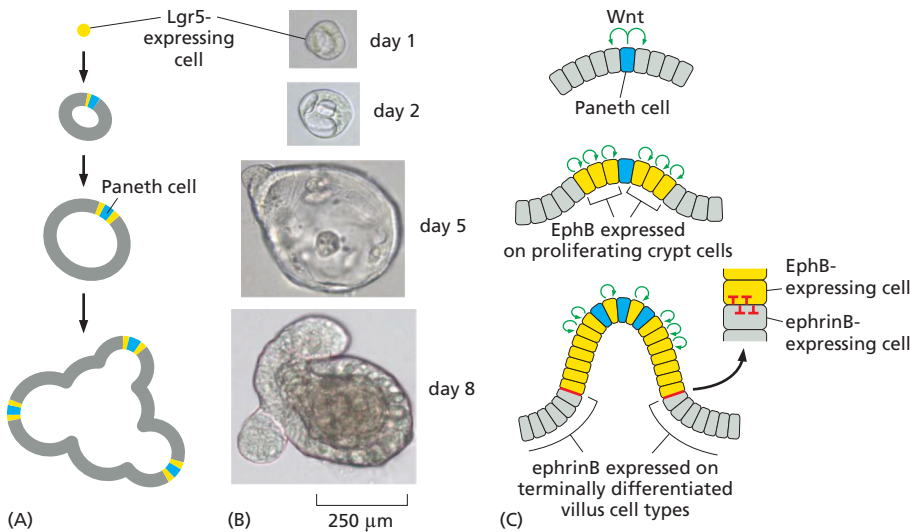


Figure 22-14 Genesis of a minigut from a single Lgr5-expressing cell cultured in a cell-free matrix. (A) Schematic drawing and (B) phase-contrast micrographs of the developing organoid. The founder cell first divides to form a small sack or vesicle. At random, one or more of the cells in this vesicle differentiates as a Paneth cell (blue). The Paneth cells secrete Wnt proteins that stimulate stem-cell self-renewal and maintain Lgr5 expression (yellow) in their immediate neighbors; besides generating more stem cells, the stem cells produce progenitor cells that differentiate into the full range of intestinal epithelial-cell types. (C) Schematic diagram showing how Paneth cell–derived Wnt signals help organize the developing crypt: besides keeping neighboring cells in the crypt proliferating in a stem-cell state, these signals activate repulsive interactions mediated by membrane-bound ephrin and Eph proteins on contacting cells (discussed in Chapter 15). These interactions cause dividing crypt cell types (which express EphB, induced by Wnt) to segregate from the nondividing, terminally differentiated villus cell types (which express ephrinB). In many tissues, cells that interact via ephrin–Eph binding repel each other when they touch (see Figure 15-52 or Figure 21-47). (Adapted from T. Sato and H. Clevers, *Science* 340:1190–1194, 2013. With permission from AAAS.)

Figure 22-4). We saw earlier that the stem cells that give rise to all the terminally differentiated cells in the intestinal epithelium express the cell-surface receptor protein Lgr5 and are located just above the base of the crypt, interspersed among the Paneth cells. The Paneth cells help create the stem-cell niche in two ways. They secrete Wnt proteins that act over a short range to stimulate stem-cell self-renewal. They also express *Delta*, a ligand for the signaling receptor Notch, on their surface, which activates Notch on the stem cells they directly contact, thereby inhibiting the cells from differentiating—an example of lateral inhibition (see Figure 15-59).

Remarkably, a single Lgr5-expressing stem cell embedded in a cell-free extracellular matrix can proliferate in a culture dish and form a tiny gut-like structure, an *organoid*, containing all the cell types normally found in gut tissue, including stem cells. **Figure 22-14** shows this process and outlines the key signaling events that establish the crypt stem-cell niche, where different cell types are produced that arrange themselves into miniature villi to form the three-dimensional organoid structure.

The Size of the Niche Can Determine the Number of Stem Cells

In each intestinal crypt, the stem-cell niche is created by 15 Paneth cells, and it only has space for a limited number of stem cells. When the stem cells divide, it is a random matter as to which of the daughter cells are pushed out of the nest; failing to get the signals they need to maintain their stem-cell identity, they are condemned to commit to differentiation. In most other stem-cell systems where this question of balance between self-renewal and commitment to differentiation has been examined, it appears that a similar mechanism operates.

The *Caenorhabditis elegans* germ-line cell lineage provides a visually striking example (**Figure 22-15**). In this system, the stem cells are maintained within a

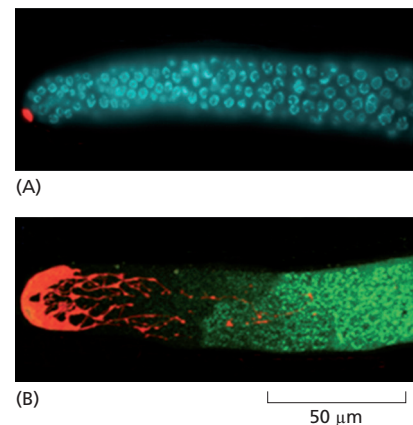


Figure 22-15 Stem-cell niche of the *C. elegans* gonad. (A) Fluorescence micrograph showing the nuclei (blue) in a portion of the *C. elegans* gonad. The nucleus of the distal tip cell is shown in red. (B) The same tissue is stained to show the cytoplasm and processes of the distal tip cell in red and the differentiated germ cells in green. The distal tip cell processes extend across multiple stem-cell diameters, maintaining stem-cell identity through Notch signaling. Those cells moved out of reach of the distal tip cell processes initiate meiosis and begin the differentiation program that will produce either eggs or sperm. (Courtesy of Judith Kimble and Sarah Crittenden.)

niche that is formed by a large somatic cell called a *distal tip cell*, which forms long, thin cellular processes. Notch ligand located on the surface of the distal tip cell signals to Notch on the stem cells to maintain their stem-cell identity. As in the intestinal crypt, the stem cells undergo self-renewal proliferation in the niche, until some of their progeny are pushed out of range of the distal tip cell processes, and, as a result, they begin to differentiate. In this way, a steady stream of cells at different stages of differentiation move away from the distal tip cell and the stem cells, as if on a conveyor belt, on their way to becoming terminally differentiated germ cells—either eggs or sperm.

Niche size, however, is not the only way to control stem-cell numbers—*asymmetric stem-cell division* is another, as we now discuss.

Asymmetric Stem-Cell Division Can Maintain Stem-Cell Number

As discussed earlier, when a stem cell divides, the two daughter cells must make a choice between remaining a stem cell or committing to differentiation. The balance between these two fates is critically important for tissue homeostasis, because excessive self-renewal would lead to an excess of stem cells, whereas excessive differentiation would deplete the stem-cell pool. We have just discussed how the size of the stem-cell niche can help control stem-cell numbers. Asymmetric stem-cell division is another mechanism for such control.

In an asymmetric stem-cell division, a process internal to the dividing stem cell causes a biased inheritance of one or more important molecules that influences the fate of the two daughter cells. For example, only one of the two daughter cells might inherit a *cell-fate determinant* that is required to maintain stem-cell identity, causing the other daughter to commit to differentiation (**Figure 22-16A**). Each such asymmetric stem-cell division produces one stem cell and one cell committed to differentiation; as a result, the number of stem cells is preserved irrespective of how many cell divisions the stem cells undergo. In symmetric stem-cell divisions, both daughters inherit such cell-fate determinants and remain stem cells, thereby increasing the pool of stem cells (**Figure 22-16B**).

As shown in Figure 22-16, prior to an asymmetric stem-cell division, the cell becomes polarized, with the cell-fate determinant localized on one side of the cell. The cell then positions the mitotic spindle to orient the plane of cell division so that only one daughter inherits the fate determinant.

Another way the orientation of the division plane in a stem cell can influence the fate of the daughter cells is to determine their position relative to the stem-cell

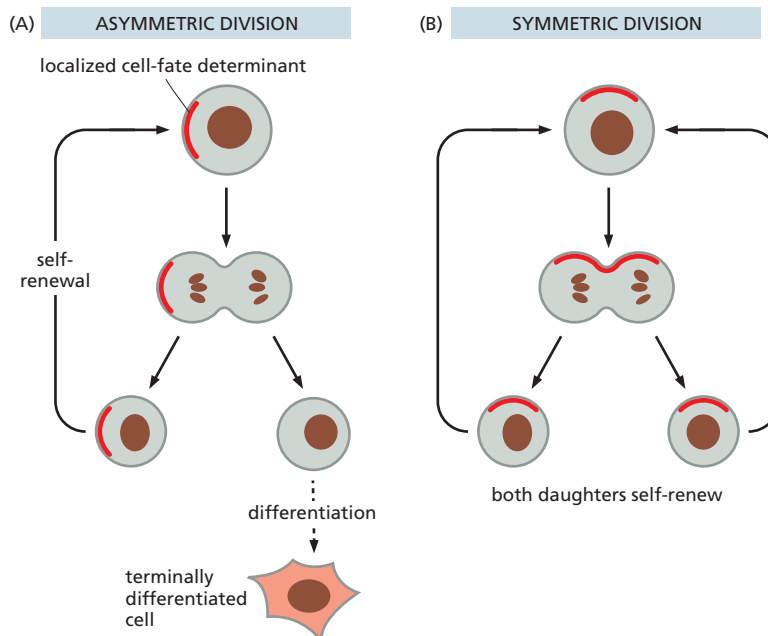


Figure 22-16 Asymmetric and symmetric stem-cell divisions. (A) In the asymmetric stem-cell division schematized here, a cell-fate determinant (red) that maintains stem-cell identity is localized to the cell cortex at one side of the cell. When the cell divides, only one daughter cell inherits the determinant and remains a stem cell, while the other daughter commits to differentiation, differentiating after a number of progenitor-cell divisions (not shown). (B) In the symmetric stem-cell division shown, both daughter cells inherit such determinants and remain stem cells.

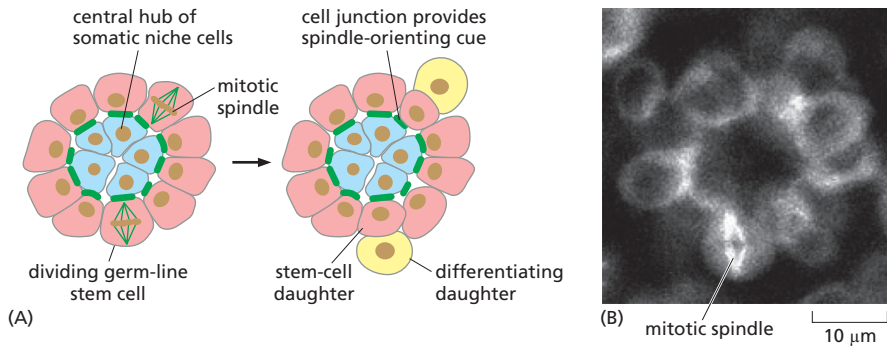


Figure 22-17 How the plane of division in a dividing, germ-line stem cell in the *Drosophila* testis determines which daughter cell maintains contact with the niche. (A) The niche consists of a hub of somatic cells, and the germ-line stem cells are arranged in a ring around the hub. When a stem cell divides, its mitotic spindle is oriented perpendicular to the niche, so that only one daughter cell maintains contact with the niche and remains a stem cell; the other daughter loses contact with the niche cells and is therefore deprived of self-renewal signals and commits to differentiation. (B) The light micrograph shows germ-line stem cells surrounding the niche and a dividing stem cell with its mitotic spindle oriented perpendicular to the niche. The stem cells have been genetically engineered to express fluorescently labeled tubulin. (B, from Y.M. Yamashita et al., *Science* 301: 1547–1550, 2003. With permission from AAAS.)

niche, such that the niche is asymmetrically inherited. In the *Drosophila* testis, for example, the germ-line stem cells orient their spindles perpendicular to the niche-supporting cells: as a result, when the stem cell divides, one daughter cell remains attached to the niche and adopts a stem-cell fate, while the other daughter is displaced away from the niche, where it is deprived of self-renewal signals and therefore commits to differentiation (Figure 22-17).

In Many Symmetric Stem-Cell Divisions, Daughter Cells Choose Their Fates Independently and Stochastically

As we have seen, asymmetric stem-cell division provides a straightforward way to maintain stem-cell numbers while generating differentiated cells in a self-renewing adult tissue. However, it leaves little room for a tissue to adapt to changing conditions, such as fluctuating nutrient availability or in response to tissue injury. In many self-renewing adult tissues, the mode of stem-cell division changes to adapt to such altered conditions. And similar adaptive changes occur during tissue development: for example, both the neural and epidermal stem cells in early developing mice mainly divide symmetrically to expand the stem-cell pool to support the rapidly growing organs; later in development, however, the stem cells shift to more asymmetric cell divisions to produce more differentiated cells (see Figure 22-16).

Stem cells in many renewing tissues, including the intestinal epithelium, adopt an even more flexible strategy. In these cases, when the stem cells divide, they do so symmetrically, and each daughter cell makes the choice between remaining a stem cell (self-renewal) and committing to differentiation, independently of its sister. In this *independent-choice mechanism*, sometimes the two daughter cells make the same choice, other times the opposite choice. The choice each cell makes might either be stochastic (probabilistic), like the flip of a coin, or governed by the environment the daughter cell finds itself in (Figure 22-18). Compared to the asymmetric-division strategy, the independent-choice mechanism is flexible, allowing local and more general environmental factors to regulate the balance of probabilities according to need—adjusting them in favor of the stem-cell option where more stem cells are required, as they often are, either for growth or for damage repair, or in favor of differentiated cells when they have been lost.

Lineage tracing experiments in the small intestine of mice reveal that some stem-cell lineages disappear while others expand and persist, as expected if the daughter cells of the stem-cell divisions make their choices stochastically (see Figure 22-18).

A Decline in Stem-Cell Function Contributes to Tissue Aging

Despite the remarkable ability of adult stem cells to persist and maintain tissue homeostasis, this ability often declines over time, contributing to tissue aging. In the case of the hematopoietic stem cells, this decline can be demonstrated experimentally using *serial transplantation*. In these experiments, the stem cells are transplanted from donor mice to irradiated host mice of the same inbred strain.

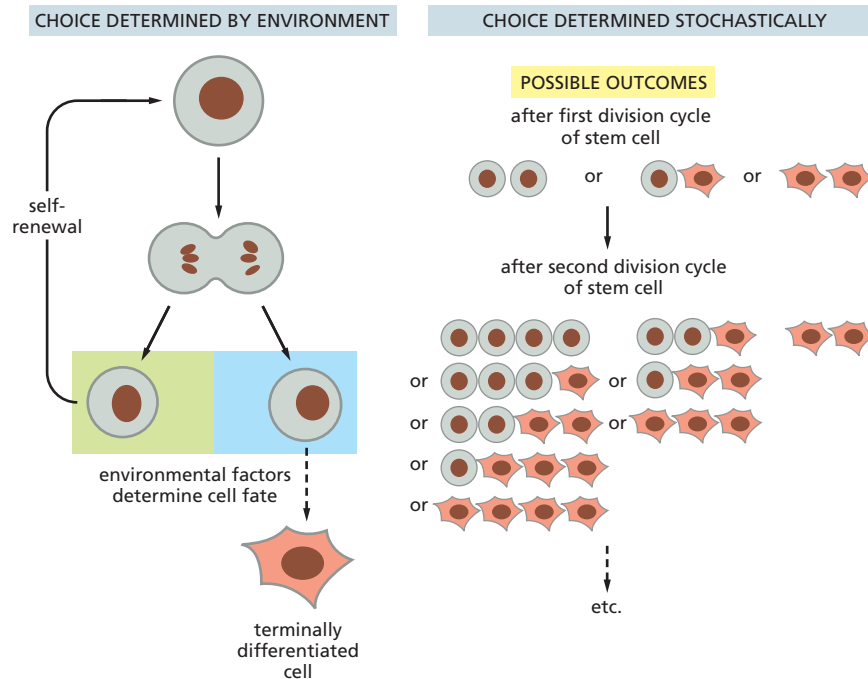


Figure 22-18 The independent-choice mechanism for how the two daughters of a stem-cell division choose their fates. The outcomes of this mechanism are more variable than that of symmetric stem-cell divisions. The choice might be determined by the local environment a daughter finds itself in (left side of the drawing) or it might be entirely stochastic (right side of drawing). With a choice made stochastically by each daughter and with a 50% probability for each one to remain a stem cell or commit to differentiation, there is, for example, a 25% chance at the first division that both daughters will commit to differentiation, so that the clone will eventually disappear. Or, at this division or later, a preponderance of daughters might choose to remain stem cells, creating a clone that persists and increases in size. With the help of some mathematics, the probability distribution of clone sizes generated from a single stem cell at any given time can be predicted on this stochastic assumption. The observations in the gut and often elsewhere fit this stochastic independent-choice strategy.

After the donor cells have repopulated the hematopoietic system of the host mice (see Figure 22-10), the hosts' hematopoietic stem cells are then transplanted into new irradiated hosts, and this sequence is repeated again and again. After multiple rounds of transplantation, the stem cells gradually lose their ability to repopulate the hematopoietic system of the new hosts, demonstrating that the stem cells change with age, apparently undergoing replicative cell senescence after repeated divisions.

To investigate whether the supportive functions of the hematopoietic stem-cell niche also decline with age, hematopoietic stem cells from young or old donor mice are transplanted into irradiated young or old host mice. Comparisons of the results from these four types of experiments (young-to-young, young-to-old, old-to-young, old-to-old—**Figure 22-19A**) indicate that a young stem-cell

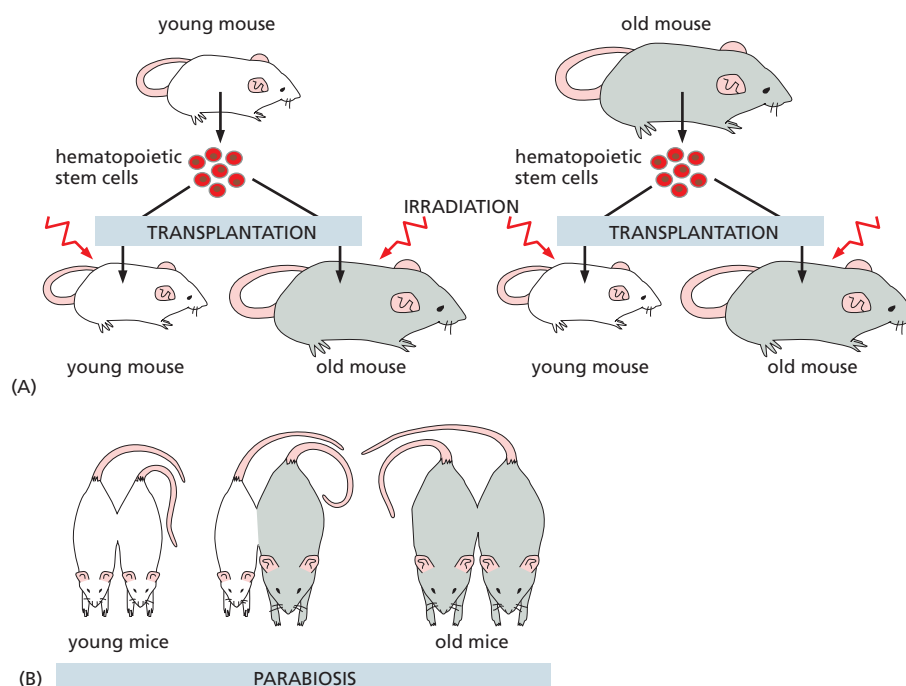


Figure 22-19 Methods to distinguish the contribution of cell-autonomous versus environmental factors to declines in hematopoietic stem-cell function during aging. (A) Hematopoietic stem cells are transplanted from young or old donor to young or old host, so that the function of stem cells can be assessed. A four-way comparison will reveal the contribution of the autonomous aging of stem cells versus the aging of the environment or niche in the decline of stem-cell function. (B) In parabiosis, animals are joined to develop a shared circulatory system. Cells in one animal will be exposed to the systemic environment of the other animal. The cellular functions in young or old partners in the heterochronic pairs (young-to-old) are compared with those in isochronic pairs (young-to-young and old-to-old). Such comparisons reveal the role of autonomous stem-cell aging compared to the role of circulating factors or the stem-cell niche in the decline of stem-cell function. [Based on M.A. Goodell and T.A. Rando, *Science* 350(6265):1199–1204, 2015, doi 10.1126/science.aab3388.]

niche cannot rescue old hematopoietic stem cells and that an old stem-cell niche can support young hematopoietic stem cells, suggesting that niche changes do not contribute to the decline of hematopoiesis with age, at least in mice.

A more general method for studying stem-cell function in aging is *parabiosis*, in which the circulatory systems of two mice of different age are connected— young-with-young, young-with-old, or old-with-old (Figure 22-19B). This three-way comparison provides a way of determining the contributions of different factors to a decline in stem-cell function with aging: the stem cells themselves, the stem-cell niche, and circulating factors in the blood. There is evidence from such studies, for example, that circulating factors from old mice can decrease neurogenesis in the hippocampus of the brain in young mice, impairing their learning and memory, mimicking the decline that occurs in the normal mouse brain during aging. Conversely, other studies find that circulating factors from young mice can restore skeletal muscle stem-cell function in old mice and can counteract age-related dysfunction in tissues including liver, pancreas, bone, and heart. Blood components that contribute to tissue aging or rejuvenation are likely to include molecules with direct actions on stem cells, and identification of these circulating factors is an active area of research.

Summary

Multiple extracellular signals operate in each stem-cell niche, enabling tissue-specific stem cells to maintain adult tissue homeostasis throughout an organism's life. These signals regulate when and where the stem cells self-renew and their daughter cells commit to differentiation. An individual stem cell's decision to self-renew or commit to differentiation is controlled to ensure the tissue maintains an adequate number of stem cells and produces an appropriate number of terminally differentiated cells. In some cases, asymmetric stem-cell divisions that produce one stem cell and one differentiating cell at each division achieve this balance. In other cases, the stem-cell divisions are symmetric, and the two daughter cells independently choose between self-renewal and commitment to differentiation, with the choice being either stochastic or in response to local environmental signals. The ability of adult stem cells to self-renew and produce terminally differentiated cells declines over time, contributing to tissue aging.

REGENERATION AND REPAIR

As we have seen, many of the tissues of the body are not only self-renewing but also self-repairing, and this is largely thanks to stem cells and the feedback controls that regulate the stem cells' behavior to maintain tissue homeostasis. There are, however, limits to what these natural repair mechanisms can achieve. In most parts of the human brain, for example, nerve cells that die, as in Alzheimer's disease, are not replaced. Likewise, when heart muscle cells die for lack of oxygen during a heart attack, they are replaced by scar tissue rather than by new heart muscle cells.

Some animals do far better than humans and can regenerate entire organs, such as whole limbs after amputation. Among the invertebrates, there are some species that can even regenerate all the tissues of the body from a single somatic cell. These phenomena encourage the hope that human cells might be coaxed by artificial measures into similar feats of repair and regeneration, to replace the nerve cells that die in individuals with Alzheimer's disease or Parkinson's disease, the insulin-secreting β cells that are lost in type 1 diabetes, the heart muscle cells that die in a heart attack, and so on. As we learn more about the basic cell biology of regeneration, these goals, once only a dream, are beginning to seem attainable.

In this section, we start with some examples of the remarkable regenerative abilities of some animal species, as an indication of what is possible in principle. We then discuss how we can improve on the natural repair processes of the human body and treat disease by exploiting the properties of the various types of stem cells found in adult human tissues.

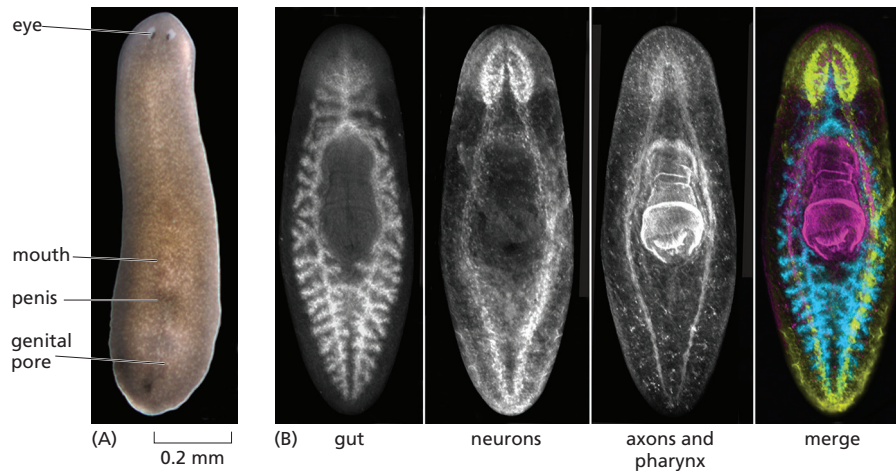


Figure 22-20 The planarian worm, *Schmidtea mediterranea*. (A) External view. (B) Immunostaining with three different antibodies, revealing the internal anatomy. (A, courtesy of A. Sánchez Alvarado; B, from A. Sánchez Alvarado, *BMC Biol.* 10:88, 2012. With permission from the author.)

Planarian Flatworms Contain Stem Cells That Can Regenerate a Whole New Body

Schmidtea mediterranea is a small freshwater flatworm, or *planarian*, a centimeter or so long when grown to full size (Figure 22-20). It has an epidermis, a gut, a brain, a pair of primitive eyes, a peripheral nervous system, musculature, and excretory and reproductive organs—most of the basic body parts familiar in other animals, although all relatively simple by vertebrate standards and built from about 20–25 distinct differentiated cell types. For more than a century, planarians such as *Schmidtea* have intrigued biologists because of their extraordinary capacity for regeneration: a small tissue fragment taken from almost any part of the body will grow and regenerate a complete new animal. This property goes with another: when the animal is starved, it gets smaller and smaller, by reducing its cell numbers while maintaining essentially normal body proportions. This behavior is called *degrowth*, and it can continue until the animal is as little as one-twentieth of its full size. Supplied with food, it will grow back to full size again. Cycles of degrowth and growth can be repeated indefinitely, without impairing survival or fertility.

Underlying this behavior is a process of continual cell turnover. Along with the differentiated cells, which do not divide, there is a population of small, undifferentiated dividing cells called *neoblasts*. The neoblasts constitute about 20% of the cells in the body and are widely distributed within it; by cell division, they serve as stem cells for the production of new differentiated cells. The differentiated cells, meanwhile, are continually dying by apoptosis, allowing their corpses to be phagocytosed and digested by neighboring cells (discussed in Chapter 18). Through this cell cannibalism, the constituents of the dying cells can be efficiently recycled. Cell division continues in a dynamic balance with cell death and cell cannibalism, no matter whether the animal is fed or starved. In conditions of starvation, the balance is tilted toward cell death and cannibalism, and in conditions of plenty, toward cell growth and division.

A high dose of x-rays halts all cell division, putting a stop to cell turnover and destroying the capacity for regeneration. The result is death after a delay of several weeks. The animal can be rescued, however, by an injection into it of a single neoblast isolated from an unirradiated donor (Figure 22-21). In a certain proportion of cases, the injected cell divides to form a clone of progeny that eventually repopulates the entire body, creating a healthy regenerative individual with an apparently complete set of terminally differentiated cell types, as well as dividing neoblasts. A gradient of positional information regarding head-to-tail identity is continually expressed by the muscle cells along the worm's body and instructs the neuroblasts' appropriate development. Genetic markers prove that the differentiated cells are all derived from the single neoblast that was injected, suggesting that at least some neoblasts are *totipotent* stem cells, in that they are able to give

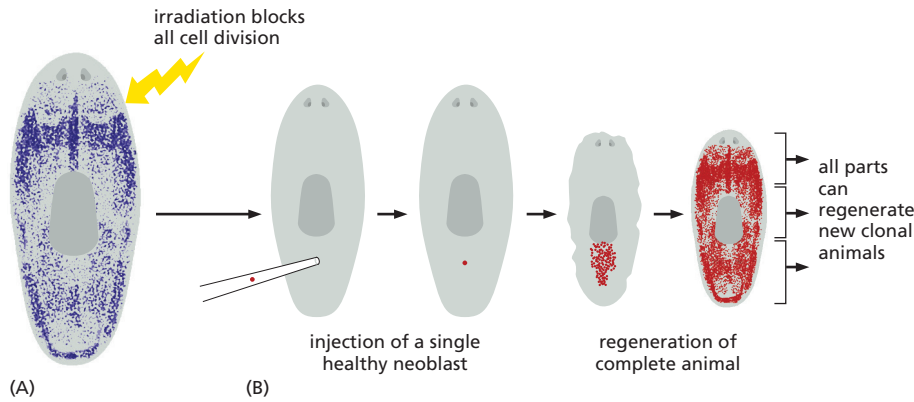


Figure 22-21 Regeneration of a planarian from a single somatic cell. (A) The distribution of dividing cells (neoblasts; blue) in the adult body. Irradiation blocks all cell division and prevents regeneration. (B) A single unirradiated neoblast cell injected into the irradiated animal is able to proliferate and reconstitute all tissues. This eventually produces a complete animal that consists entirely of the progeny of this one cell and can regenerate normally. A gradient of information concerning head-to-tail identity is continually expressed in muscle cells along the body and instructs the neuroblasts. (Adapted from E.M. Tanaka and P.W. Reddien, *Dev. Cell* 21:172–185, 2011.)

rise to all the cell types that make up the body of the flatworm, including more neoblasts like themselves.

Some Vertebrates Can Regenerate Entire Limbs and Organs

One might think that such powers of regeneration would be a prerogative of only small, simple, primitive animals. But some vertebrates, too, especially fish and amphibians, show remarkable regenerative abilities. Salamanders, including newts and axolotls, for example, can regenerate a whole or a part of an amputated limb, as well as many other body parts, including brain tissue and spinal cord (**Figure 22-22**). In the process of limb regeneration, a *blastema*—a small bud resembling an embryonic limb bud—forms at the site of amputation. The rapid lateral migration of adjacent epidermal cells serves to seal the wound site. The underlying blastema cells are mostly derived from activated stem cells and progenitor cells in the stump that are highly proliferative. Although the cells all look alike, they retain a memory of their tissue of origin and enough positional information to enable them to differentiate into the appropriate cell types and to form a correctly patterned replacement for the limb or the part of the limb that was amputated. The regenerative process looks like a recapitulation of embryonic limb development. As in the developing limb, all of the required cell behaviors are orchestrated by intercellular signals that are produced by the overlying epidermis, the ingrowing nerves, the proliferating lineage-restricted stem cells, and the transit-amplifying progenitor cells and their differentiated

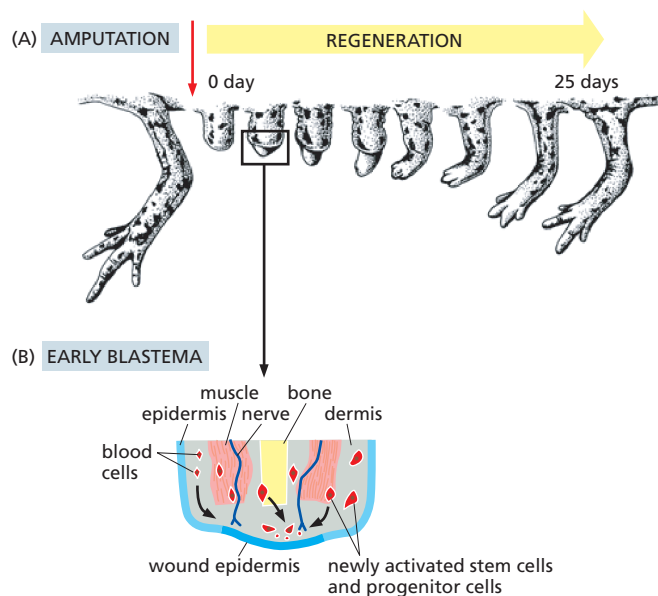


Figure 22-22 Newt limb regeneration. (A) The time-lapse sequence shows the stages of regeneration after amputation at the mid-humerus level. The sequence spans the events of wound healing, activation and proliferation of stump stem cells and progenitor cells in the process of blastema formation, and differentiation of various limb-cell types. (B) Schematic diagram of the early blastema. (A, courtesy of Susan Bryant and David Gardiner.)

progeny. Most of the signaling pathways involved are the same as those used in limb development, including those activated by Hedgehog, Wnt, and FGF family members.

Why a salamander can regenerate so many body parts whereas a mammal cannot remains a profound mystery.

Stem Cells Can Be Used Clinically to Replace Lost Hematopoietic or Skin Cells

Earlier in this chapter, we saw how mice can be irradiated to kill off their hematopoietic stem and progenitor cells and then rescued by a transfusion of new stem cells, which repopulate the bone marrow and restore blood-cell production (see Figure 22-10). In the same way, human individuals with some forms of leukemia or lymphoma can be irradiated or chemically treated to destroy their cancer cells, along with the rest of their hematopoietic tissue, and then can be rescued by a transfusion of healthy, noncancerous hematopoietic stem cells. In favorable cases, the stem cells can be sorted out from samples of an individual's own hematopoietic tissue before it is ablated; the stem cells are then transfused back afterward, thereby avoiding problems of immune rejection.

Another example of the use of stem cells is in the repair of the skin after extensive burns. By culturing cells from undamaged regions of the burned person's skin, it is possible to generate enough epidermal stem cells by long and complicated procedures to repopulate the damaged body surface.

Neural Stem Cells Can Be Manipulated in Culture and Used to Repopulate a Diseased Central Nervous System

The vertebrate central nervous system (the CNS) is the most complex tissue in the body, at an opposite extreme from the epidermis. And yet fish and amphibians can regenerate large parts of their brain, spinal cord, and eyes after these have been cut away. In adult mammals, however, these tissues have very little capacity for self-repair, and stem cells capable of generating new neurons are hard to find—so hard to find that for many years they were thought to be absent.

We now know, however, that neural stem cells that generate neurons, glial cells, or both, do persist in certain parts of the adult mammalian brain, as shown in **Figure 22-23** for stem cells that give rise to neurons in the mouse olfactory bulb. In both the adult mouse and human brain, there is also a continual turnover of neurons in the hippocampus, a region specifically concerned with learning and memory. Here, plasticity of adult brain function is associated with turnover of a particular subset of neurons: about 1400 new neurons in this class are generated every day, which is a turnover rate of 1.75% of the cells per year.

A more dramatic neuronal turnover is observed in the brains of certain songbirds. In these birds, large numbers of neurons die each year and are replaced by new neurons, as part of a process by which the birds refine their song for each new breeding season.

Brain neural stem cells can be studied in culture. Fragments taken from self-renewing regions of the adult or fetal mammalian brain, for example, can be dissociated and cultured under conditions where they form floating “neurospheres”—clusters consisting of a mixture of neural stem cells and their neuronal and glial progeny cells. The neurospheres can be propagated through many cell generations, and their cells can be taken at any time and implanted back into the brain of an intact animal, where they will produce differentiated neurons and glial cells.

Using slightly different culture conditions, with the right combination of growth factors in the culture medium, the neural stem cells can be grown as dissociated cells in a culture dish and induced to proliferate as an almost pure stem-cell population without attendant differentiated progeny. By a further change in the culture conditions, these cells can be induced at any time to differentiate to give

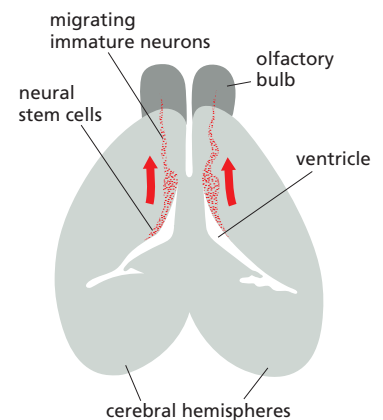
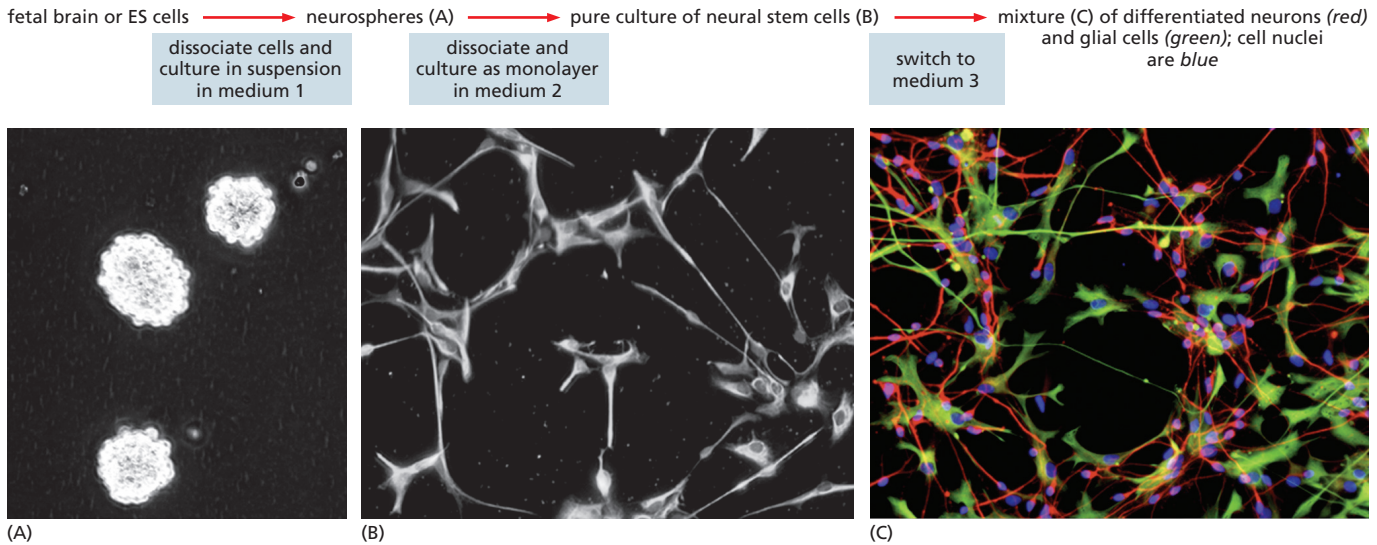


Figure 22-23 The continuing production of neurons in an adult mouse brain.

The brain is viewed from above, in a cutaway section, to show the region lining the ventricles of the forebrain where neural stem cells are found. These cells continually produce progeny that migrate to the olfactory bulb, where they terminally differentiate into neurons. The constant turnover of neurons in the olfactory bulb is presumably linked in some way to the turnover of the olfactory receptor neurons that project to it from the olfactory epithelium in the nose. In adult mice and humans, there is also a continuing turnover of neurons in the hippocampus (not shown), a region specially concerned with learning and memory. (Adapted from B. Barres, *Cell* 97:667–670, 1999.)



either a mixture of neurons and glial cells (**Figure 22-24**) or just one of these two classes of cells, according to the composition of the medium.

Neural stem cells, whether derived as above or from the pluripotent stem cells that we describe in the final section of this chapter, can be grafted into an adult brain. Once there, they show a remarkable ability to adjust their behavior to match their new location. Stem cells from the mouse hippocampus, for example, when implanted into the migration pathway of the mouse olfactory-bulb precursor cells (see **Figure 22-23**), give rise to neurons that become correctly incorporated into the olfactory bulb. This capacity of neural stem cells and their progeny to adapt to a new environment in animals suggests applications in the treatment for central nervous system diseases or injury.

Summary

Animals vary in their capacity for regeneration. At one extreme, planarian flatworms contain stem cells (neoblasts) that support the continual turnover of all cell types, and an entire worm can be regenerated from practically any small body fragment or even from a single neoblast cell. Salamanders can regenerate limbs and other large body parts after amputation, but the regenerating cells remain restricted according to their origins: muscle cells in the regenerate derive from muscle, epidermis from epidermis, and so on. In mammals, regeneration is more limited. Nevertheless, it is becoming possible to go beyond the natural limits of wound healing by exploiting stem-cell biology. Certain regions of the nervous system contain stem cells that support production of neurons in these sites throughout life. Neural stem cells can be obtained from these sites or from fetal brains, grown in culture, and then grafted back into other sites in the brain, where they are able to generate neurons appropriate to the new location.

CELL REPROGRAMMING AND PLURIPOTENT STEM CELLS

When cells are transplanted from one site in the mammalian body to another or are removed from the body and maintained in culture, they remain largely faithful to their origins. Each type of specialized cell has a memory of its developmental history and seems fixed in its fate. Some limited transformations can certainly occur, and, as we have seen, some stem cells can generate a variety of differentiated cell types, but the possibilities are restricted. Each type of adult stem cell serves for the renewal of one particular type of tissue, and the whole pattern of self-renewing stem cells to differentiated cells in the adult body is amazingly stable. What, at a fundamental molecular level, is the nature of these stable differences between cell types and cell states? Is there any way to override

Figure 22-24 Neural stem cells. Shown are the steps leading from fetal brain tissue or embryonic stem cells, via neurospheres (A), to a pure culture of neural stem cells (B). These stem cells can be kept proliferating as such indefinitely or, through a change of medium, can be caused to differentiate (C) into neurons (red) and glial cells (green). Neural stem cells with the same properties can also be derived, via a similar series of steps, from embryonic stem (ES) or induced pluripotent stem (iPS) cells (discussed later in this chapter). (Micrographs from L. Conti et al., *PLoS Biol.* 3:e283, 2005. With permission from the authors.)

the cell-memory mechanisms and force a switch from one state to another that is radically different?

We have already discussed these fundamental questions from a general standpoint in Chapter 7. Here we consider them more closely in the context of stem-cell biology, where there has been a recent revolution in our understanding and in our ability to manipulate states of cell differentiation. With further research, these advances will likely have many important medical applications.

Nuclei Can Be Reprogrammed by Transplantation into Foreign Cytoplasm

If we cannot switch the basic character of a specialized cell by changing its environment, can we do so by interfering with its inner workings in a more direct and drastic way? An extreme treatment of this sort is to take the nucleus of the cell and transplant it into the cytoplasm of a large cell of a different type. If the factors that define and maintain a particular cell type are in the cytoplasm, the transplanted nucleus should switch its pattern of gene expression to conform with that of the host cell. In Chapter 7, we described a famous experiment of this sort, using the frog *Xenopus*. In this experiment, the nucleus of a differentiated cell (a cell from the lining of a tadpole's gut) was used to replace the nucleus of an oocyte (an egg-cell precursor arrested in prophase of the first meiotic division, in readiness for fertilization). The resulting hybrid cell went on, in a certain fraction of cases, to develop into a completely normal frog (see Figure 7-2A). This was crucial evidence for what is now a central principle of developmental biology: the cell nucleus, even that of a differentiated cell, contains a complete genome, capable of supporting development of all the normal cell types of the organism. At the same time, the experiment showed that cytoplasmic factors can indeed reprogram a nucleus: the oocyte cytoplasm can drive the gut-cell nucleus back to an early embryonic state, from which it can then step through the changing patterns of gene expression that lead all the way to a complete adult organism.

The full story, however, is not quite so simple. First, the reprogramming in such experiments is not perfect. When the transplanted nucleus is taken from a gut cell, for example, a gene that is normally specific to the gut is found to be expressed persistently, even in the muscle cells of the final animal. Second, the experiment succeeds in only a limited proportion of cases, and this success rate becomes lower and lower, the more mature the animal from which the transplanted nucleus is taken: very large numbers of transplantations must be performed to achieve a single success if the nucleus comes from a differentiated cell of an adult frog.

Nuclear transplantation can be done in mammals too, with basically similar results. For example, a nucleus taken from a differentiated cell in the mammary gland of an adult sheep and transplanted into an enucleated sheep's egg was able to support development of an apparently normal sheep—the famous Dolly. Again, the success rate is low: many transplantations have to be done to obtain one such individual.

Reprogramming of a Transplanted Nucleus Involves Drastic Changes in Chromatin

In a typical fully differentiated cell, there seem to be mechanisms maintaining the pattern of gene expression that cytoplasmic factors cannot easily override. An obvious possibility is that the stability of the pattern of gene expression in an adult cell may depend, in part at least, on self-perpetuating modifications of chromatin, as discussed in Chapter 4 (see Figure 4-44). As explained in Chapter 7, the phenomenon of X-inactivation in mammals provides a clear example of such epigenetic control. Two X chromosomes are present in each female cell, exposed to the same chemical environment, but while one remains active, the other persists from one cell generation to the next in a condensed inactive state; cytoplasmic factors cannot be responsible for the difference, which must instead reflect

mechanisms intrinsic to the individual chromosome. Elsewhere in the genome also, controls at the level of chromatin act in combination with other forms of regulation to govern the expression of each gene. Genes can be shut down completely, or switched on constitutively, or maintained in a labile state where they can be readily switched on or off according to changing circumstances.

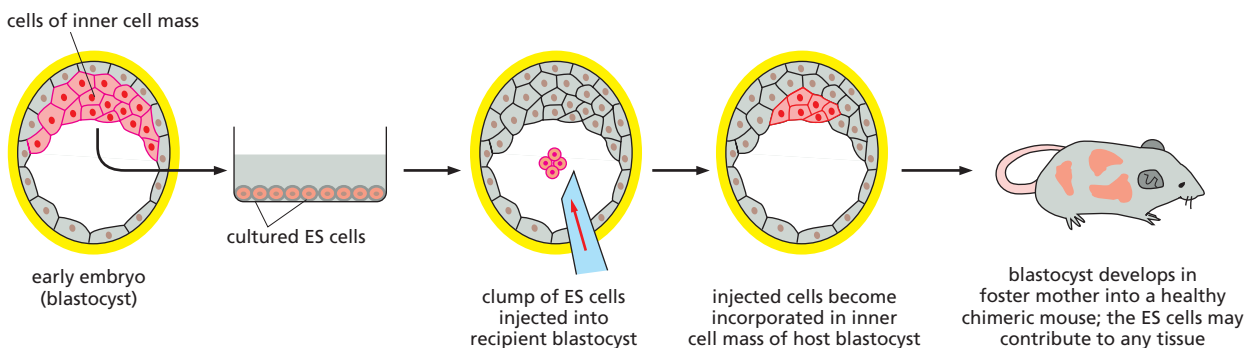
The reprogramming of a nucleus transplanted into an oocyte involves dramatic changes in chromatin. The nucleus swells, increasing its volume 50-fold as the chromosomes decondense; there is a wholesale alteration in patterns of methylation of DNA and histones; the linker histone H1 (the histone that links adjacent nucleosomes) is replaced by a variant form that is peculiar to the oocyte and early embryo; and the preexisting type of histone H3 is also replaced at many sites by a distinct isoform. Evidently, the egg contains factors that reset the state of the chromatin in the nucleus, wiping out old histone modifications on chromatin and imposing new ones. Reprogrammed in this way, the genome becomes competent once again to initiate embryonic development and to give rise to the full range of differentiated cell types.

Embryonic Stem (ES) Cells Can Generate Any Part of the Body

A fertilized egg, or an equivalent cell produced by nuclear transplantation, is a remarkable thing: it can generate a whole new multicellular individual, which means that it can give rise to every normal type of specialized cell, including even egg or sperm cells for production of the next generation. A cell in such a state is said to be **totipotent**; a cell that can give rise to most cell types but not absolutely all is said to be **pluripotent**. Nevertheless, such a totipotent or pluripotent cell is not a stem cell as it is not self-renewing, but is instead dedicated to a program of progressive differentiation. If it were the only available starting point for study and exploitation of pluripotent cells, the enterprise would require a continual supply of fresh fertilized eggs or fresh nuclear transplantation procedures—an awkward requirement for studies in experimental animals, and unacceptable for practical applications in humans.

Here, however, nature has been unexpectedly kind to scientists. It is possible to take an early mouse embryo, at the blastocyst stage, and through cell culture to derive from it a class of stem cells called **embryonic stem cells**, or **ES cells**. ES cells originate from the inner cell mass of the early embryo (the cluster of cells that give rise to the body of the embryo proper, as opposed to extraembryonic structures), and they have an extraordinary property: given suitable culture conditions, they will continue proliferating indefinitely and yet retain an unrestricted developmental potential. Their only limitation is that they do not give rise to extraembryonic tissues such as those of the placenta. Thus they are classified as pluripotent, rather than totipotent. But this is a minor restriction. If ES cells are put back into a blastocyst, they become incorporated into the embryo and can give rise to all the tissues and cell types in the body, integrating perfectly into whatever site they may come to occupy, and adopting the character and behavior that normal cells would show at that site (Figure 22–25). They can even give rise to germ cells, from which a new generation of animals can be derived.

Figure 22–25 Production and pluripotency of ES cells. ES cells are derived from the inner cell mass (ICM) of the early embryo. The ICM cells are transferred to a culture dish containing an appropriate medium, where they become converted to ES cells and can be kept proliferating indefinitely without differentiating. The ES cells can be taken at any time—after genetic manipulation, if desired—and injected back into a developing blastocyst. There they incorporate into the inner cell mass and take part in formation of a well-formed chimeric animal that is a mixture of ordinary and ES-derived cells. The ES-derived cells can differentiate into any of the cell types in the body, including germ cells from which a new generation of mice can be produced, which are no longer chimeric, but consist of cells that all inherit half their genes from the cultured ES cell line.



ES cells let us move between cell culture, where we can use powerful techniques for genetic transformation and selection, and the intact organism, where we can discover how such genetic manipulations affect development and physiology. Thus, ES cells opened the way to efficient genetic engineering in mammals, leading to a revolution in our understanding of mammalian molecular and developmental biology.

Cells with properties similar to those of mouse ES cells can also be derived from early human embryos and from human fetal germ cells, and even, as we explain shortly, from differentiated cells taken from adult mammalian tissues. In this way, one can obtain a potentially inexhaustible supply of pluripotent cells. Grown in culture, these cells can be manipulated, by suitable choice of culture conditions, to give rise to large quantities of almost any type of differentiated cell, opening the way to many practical applications. Before discussing them, however, we consider the underlying biology.

A Core Set of Transcription Regulators Defines and Maintains the ES-Cell State

What is it that gives ES cells and related types of pluripotent stem cells their extraordinary developmental potential? And what can they tell us about the fundamental mechanisms underlying stemness, commitment to differentiation, and the stability of the differentiated state?

For some ES-cell attributes, the answer is simple. For example, an essential feature of ES cells is that they must avoid *replicative cell senescence*. As discussed in Chapter 17, this is the fate of fibroblasts and many other types of proliferating somatic cells: such cells are limited in the number of times they will divide, in part at least because they lack telomerase activity, with the result that their telomeres become shorter with each division cycle, leading eventually to a permanent cell-cycle arrest. ES cells, by contrast, express high levels of active telomerase, allowing them to escape replicative cell senescence and continue to divide indefinitely. This is a property shared with other, more developmentally restricted types of stem cells, such as those of the adult intestine, which similarly can carry on dividing for hundreds or thousands of cell cycles.

The deeper problem is to explain how the whole complex pattern of gene expression in an ES cell is organized and maintained. As a first step, one can look for genes expressed specifically in ES cells or in the corresponding pluripotent cells of the early embryo. This approach identifies a relatively small number of candidate *ES-critical genes*; that is, genes that seem to be essential in one way or another for the peculiar character of ES cells. A gene called *Oct4*, for example, is exclusively expressed in ES cells and in related classes of cells in the intact organism—specifically, in the germ-cell lineage and in the inner cell mass and its precursors. *Oct4* codes for a transcription regulator. When it is lost from ES cells, they lose their ES-cell character, and when it is missing in an embryo, the cells that should specialize as inner cell mass are diverted into an extraembryonic pathway of differentiation, and the embryo's development is aborted.

Fibroblasts Can Be Reprogrammed to Create Induced Pluripotent Stem (iPS) Cells

In Chapter 7, we saw that fibroblasts and some other cell types can be driven to switch their character and differentiate as muscle cells if the master muscle-specific transcription regulator MyoD is artificially expressed in them. Could the same technique be used to convert fibroblasts and other cell types into ES cells, through forced expression of ES-critical genes such as *Oct4*? This question was tackled by transfecting mouse fibroblasts with retroviral vectors carrying genes that one might hope to have such an effect. A total of 24 candidate ES-critical genes were tested in this way. None of them was able by itself to cause the conversion, but in certain combinations they could do so. In 2006, the first breakthrough experiments whittled down the requirement to a core set of four

genes, all of them encoding transcription regulators—Oct4, Sox2, Klf4, and Myc, known as the OSKM factors for short. When co-expressed, these could reprogram mouse fibroblasts, permanently converting them into cells very similar to ES cells (Figure 22–26). ES-like cells created in this way are called **induced pluripotent stem cells**, or **iPS cells**. Like ES cells, iPS cells can continue dividing indefinitely in culture, and when incorporated into a mouse blastocyst they can participate in creation of a perfectly formed chimeric animal. In this animal, they can contribute to the development of any tissue and can turn into any differentiated cell type, including functional germ cells from which a new generation of mice can be raised (see Figure 22–25).

iPS cells can now be derived from adult human cells, including from various differentiated cell types besides fibroblasts. Numerous methods can be used to drive expression of the transforming OSKM factors, including methods that leave no trace of foreign DNA in the reprogrammed cells. Variations of the original cocktail of transcription regulators can drive the conversion, with different specialized cell types having somewhat different requirements. Myc overexpression, for example, turns out not to be absolutely necessary, although it enhances the efficiency of the process. And differentiated cell types may express some of the required factors as part of their normal phenotype. For example, certain cells of hair follicles already express Sox2, Klf4, and Myc; to convert them into iPS cells, it is enough to force them artificially to express Oct4.

Reprogramming Involves a Massive Upheaval of the Gene Control System

Converting a differentiated cell into an iPS cell is not like flicking a switch on some predictable, precisely engineered piece of machinery. Only a few of the cells that receive the OSKM factors will actually become iPS cells—one in several thousand in the original experiments, and still only a small minority with more recent, improved techniques. In fact, the success of the original experiments depended on clever selection strategies to pick out those few cells where the conversion had occurred (Figure 22–27).

Conversion to an iPS state by the OSKM factors is not only inefficient but also slow: fibroblasts take 10 days or more from introduction of the conversion factors before they begin to express markers characteristic of iPS cells. This suggests the transformation involves a long cascade of changes. These changes have been extensively studied, and they affect both the expression of individual genes and the state of the chromatin. The time course is outlined in Figure 22–28. The process begins with a Myc-induced cell proliferation and loosening of chromatin structure that promotes the binding of the other three transcription regulators to many hundreds of different sites in the genome. At a large proportion of these sites, Oct4, Sox2, and Klf4 all bind in concert. The binding sites include the endogenous *Oct4*, *Sox2*, and *Klf4* genes themselves, which eventually creates positive feedback loops like those just described, making expression of these genes self-sustaining (see Figure 22–26). But self-induction

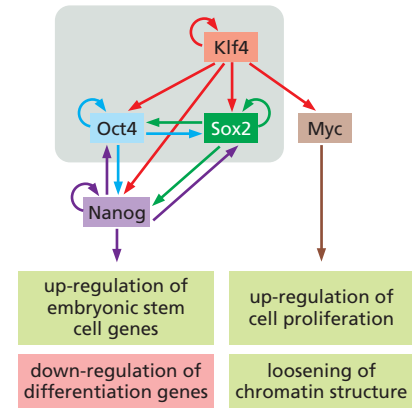


Figure 22–26 Reprogramming fibroblasts to iPS cells with the OSKM factors.

As indicated, the transcription regulatory proteins Oct4, Sox2, and Klf4 (OSK factors) induce both their own and each other's synthesis (gray shading). This generates a self-sustaining feedback loop that helps to maintain cells in an ES cell-like state, even after all of the experimentally added OSKM initiators have been removed. Myc overexpression speeds up early stages of the reprogramming process through the mechanisms shown (see Figure 17–59). Stable reprogramming also involves the permanently induced expression of the *Nanog* gene, which encodes an additional transcription regulator (see Figure 7–10). (Adapted from J. Kim et al., *Cell* 132: 1049–1061, 2008.)

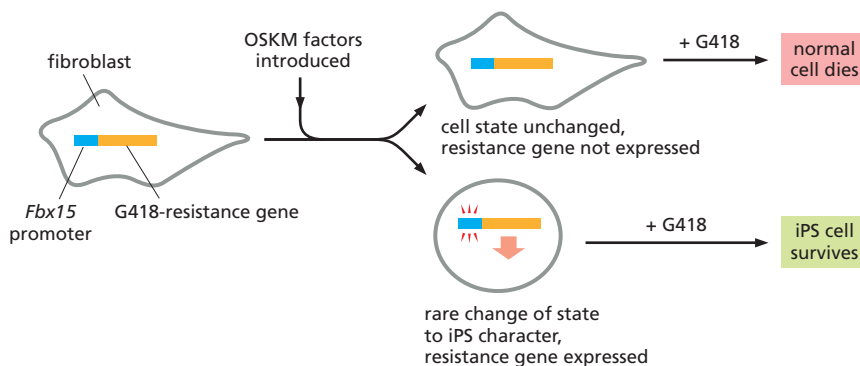


Figure 22–27 A strategy used to select cells that have converted into iPS cells.

The experiment makes use of a gene (*Fbx15*) that is present in all cells but is normally expressed only in ES and early embryonic cells (although not required for their survival). G418 is an aminoglycoside antibiotic that blocks protein synthesis in both bacteria and eukaryotic cells. A fibroblast cell line is genetically engineered to contain a gene that produces an enzyme that degrades G418 under the control of the *Fbx15* regulatory sequence. When the OSKM factors are artificially expressed in this cell line, a small proportion of the cells undergo a change of state and activate the *Fbx15* regulatory sequence, driving expression of the G418-resistance gene. When G418 is added to the culture medium, these are the only cells that survive and proliferate. When tested, they turn out to have iPS-cell characteristics.

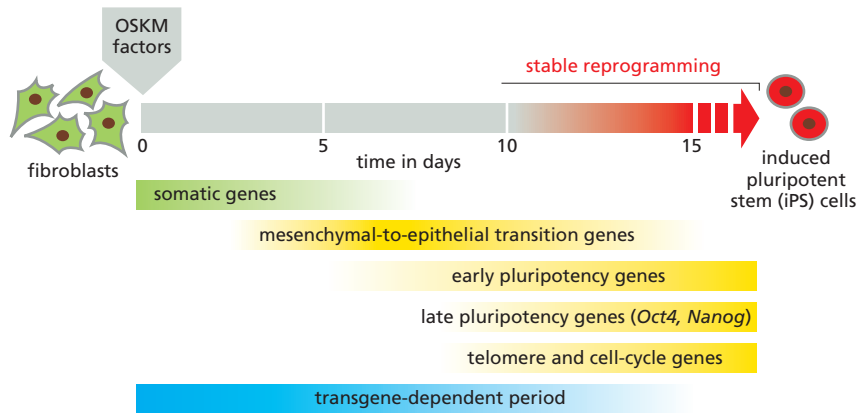


Figure 22-28 A summary of some of the major events that accompany the reprogramming of fibroblasts to iPS cells. Expression of the OSKM factors induces a series of events over a period of days to weeks. Reprogramming begins with down-regulation of somatic-cell markers. Morphological changes characteristic of a *mesenchymal-to-epithelial transition* ensue (see p. 1197), driven in part by changes in the expression of cell adhesion and signal proteins. Induction of early pluripotency markers is followed by the expression of pluripotency genes such as *Nanog* and *Oct4*. With similar timing, cells become immortalized as telomerase is induced and cell-cycle genes are regulated to enable stem-cell self-renewal. Stable reprogramming occurs in the time window when cells activate endogenous pluripotency genes and become independent of the OSKM factors. Notably, the vast majority of fibroblasts expressing the OSKM factors fail to down-regulate somatic markers and activate pluripotency genes and do not convert to iPS cells. (Adapted from M. Stadtfeld et al., *Cell Stem Cell* 2:230–240, 2008. With permission from Elsevier.)

of *Oct4*, *Sox2*, and *Klf4* is only a small part of the transformation that occurs. These three core factors activate some target genes and repress others, producing a cascade of effects that reorganize the gene control system globally and at every level, changing the patterns of histone modification, DNA methylation, and chromatin compaction, as well as the expression of innumerable proteins and noncoding RNAs. By the end of this complex process, the resulting iPS cell is no longer dependent on the artificially generated OSKM factors that triggered the change: it has settled into a stable, self-sustaining state of coordinated gene expression, making its own OSKM factors (and all the other essential ingredients of a pluripotent stem cell) from its own endogenous copies of the genes.

An Experimental Manipulation of Factors That Modify Chromatin Can Increase Reprogramming Efficiencies

The low efficiency and slow rate of conversion in early iPS-cell studies suggested that there are barriers that normally block the switch from the differentiated state to the iPS state in these experiments, and that overcoming them can be a difficult process that involves a large element of chance. This probably helps explain why the outcome is often variable, with significant differences between the individual iPS cell lines generated, even when the initial differentiated cells are genetically and phenotypically identical. Only some of the candidate iPS lines produced have passed all the tests of pluripotency. Moreover, at a molecular level, there are differences even among the fully validated iPS lines: although they share many features, they vary in details of their gene expression patterns and, for example, in their patterns of DNA methylation.

Overcoming these difficulties is critical for improving our understanding of how cell specialization is controlled and organized in multicellular organisms; it should also facilitate many medical advances. Thus, intensive research is being carried out on the reprogramming process. One approach aims at obtaining a much clearer picture of the role that chromatin structures play in gene regulation in eukaryotes.

From our discussion of nuclear transplantation, one might expect that any reprogramming of a differentiated cell would require a radical and widespread change in the chromatin structure associated with selected genes. Not only are such changes observed, but a large number of different experiments reveal that the efficiency of the reprogramming process can be substantially increased by altering the activity of proteins that affect chromatin structure. **Figure 22-29** categorizes some of the factors that when manipulated can enhance the transformation of fibroblasts to iPS cells; those in the top three rows—chromatin remodelers, histone modifiers, and histone variants—are especially well known to have profound effects on the organization of nucleosomes in chromatin (discussed in Chapter 4).

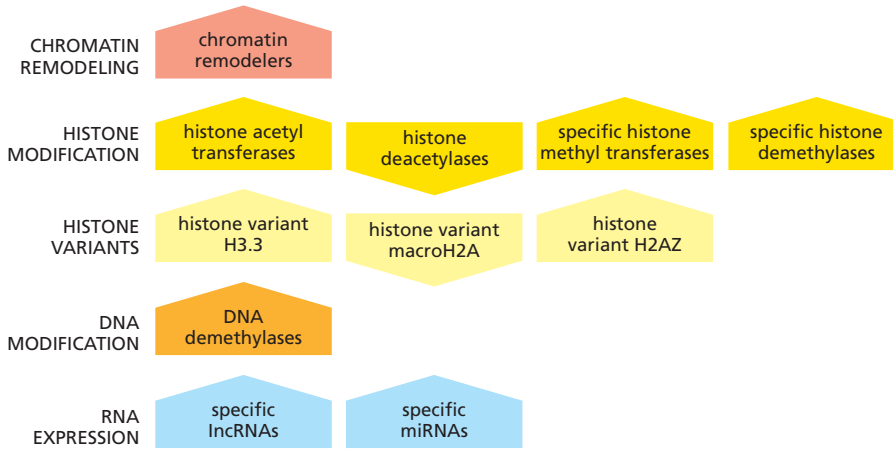


Figure 22–29 Factors that have been observed to enhance reprogramming efficiency. Emphasized here are those factors that can alter chromatin states, with those in the top three rows having the most direct effects. An *up arrow* indicates that reprogramming is increased when the activity of the indicated factor is increased; a *down arrow* indicates that reprogramming is increased when the activity of the indicated factor is decreased. Thus, for example, increased activity of histone acetyl transferases and increased activity of histone deacetylases have opposite effects, as expected from their biochemical activities (see p. 206). Note that histone chaperones also contribute to programming efficiency and may do so either positively or negatively depending on the histone variants assembled into chromatin.

ES and iPS Cells Can Be Guided to Generate Specific Adult Cell Types and Even Organoids

We can think of embryonic development in terms of a series of choices presented to cells as they follow a road that leads from the fertilized egg to terminal differentiation. After their long sojourn in culture, the ES cells or iPS cells and their progeny can still read the signs at each branch in the highway and respond as normal early embryonic cells would. If ES or iPS cells are implanted directly into an embryo at a later stage of development or into an adult tissue, however, they fail to receive the appropriate sequence of cues; their differentiation then is not properly controlled, and they will often give rise to a tumor called a *teratoma*, containing a mixture of cell types inappropriate to the site in the body (**Figure 22–30**).

By exposing ES or iPS cells in culture to an appropriate sequence of growth factors and other signal proteins, delivered with the right timing, it is possible to guide the cells along a pathway that approximates a normal developmental pathway and convert them into standard, differentiated, adult cell types (**Figure 22–31** and **Movie 22.2**). Success requires trial and error but has now been achieved for many differentiated mouse and human cell types, including human dopaminergic neurons and human insulin-producing β cells, the cells lost in Parkinson's disease and type 1 diabetes, respectively.

Remarkably, under appropriate conditions, mouse or human ES and iPS cells and their progeny can proliferate, differentiate, and self-assemble in culture to form miniature, three-dimensional organs called **organoids**, which closely resemble the normal organ in its organization. An early striking example is shown in **Figure 22–32**, where a developing eye organoid is formed from human ES cells.

Mouse and human ES and iPS cells, and progenitor cells derived from them, have been used to form organoids that resemble a large variety of developing organs, including important parts of the human brain, arguably the most complex and sophisticated structure on Earth. Such organoids provide powerful models for studying organ development in a culture dish, where one can identify and study the genes involved and explore the roles of cell–cell interactions in ways not possible in the intact organism. We discussed earlier the remarkable demonstration that a single multipotent intestinal stem cell can form a complex intestinal organoid under appropriate conditions in culture (see **Figure 22–14**).

Cells of One Specialized Type Can Be Forced to Transdifferentiate Directly into Another

The route we have just described, from one type of differentiated cell to another type of differentiated cell via conversion to an iPS cell, seems needlessly roundabout. Could we not convert differentiated cell type A into differentiated cell type B directly, without backtracking to a pluripotent stem cell? For many years, it has been known that such *transdifferentiation* can be achieved in a few special cases,



Figure 22–30 A teratoma. These tumors contain tissues derived from each of the embryonic germ layers: endoderm, mesoderm, and ectoderm (including hair and teeth, as seen here). The tumors are usually benign and can develop when ES or iPS cells are injected into an immunocompatible adult animal, demonstrating the pluripotency of these stem cells. (Courtesy of Cao Xuan Cu.)

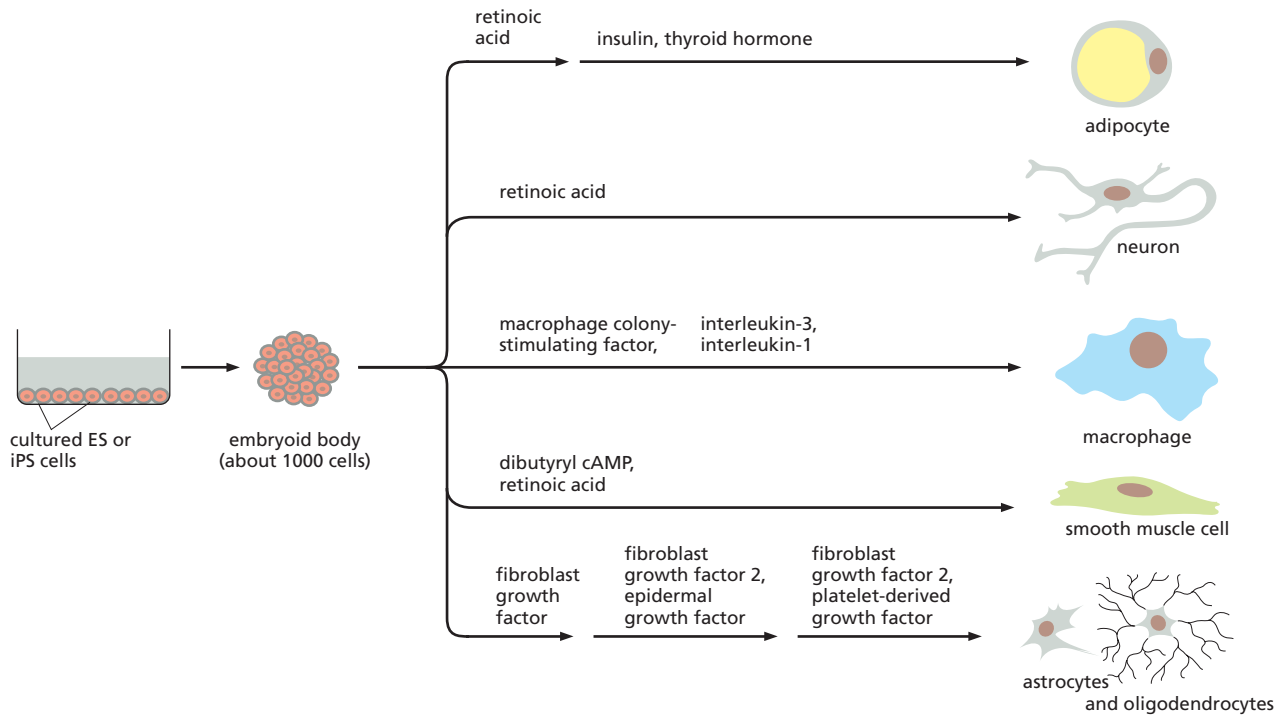


Figure 22–31 Production of differentiated cells from mouse or human ES or iPS cells in culture. ES and iPS cells can be cultured indefinitely as pluripotent cells when attached as a monolayer to a dish. Alternatively they can be detached and allowed to form aggregates called *embryoid bodies*, which causes the cells to begin to specialize. Cells from embryoid bodies, cultured in media with different factors added, can then be driven to differentiate in various ways. Notably, it takes very much longer to generate the differentiated cells from human compared to mouse pluripotent stem cells, reflecting the much slower rate of human development compared to a mouse. (Based on E. Fuchs and J.A. Segre, *Cell* 100:143–155, 2000.)

such as the conversion of fibroblasts into skeletal muscle cells by forced expression of MyoD (see p. 428). But now, with the insights that have come from the study of ES and iPS cells, ways have been found to bring about such interconversions in a much wider range of cases, including conversion of fibroblasts into neurons, hepatocytes, and intestinal epithelial cells.

An elegant example with special medical relevance comes from studies of the heart. By forcing expression of an appropriate combination of transcription

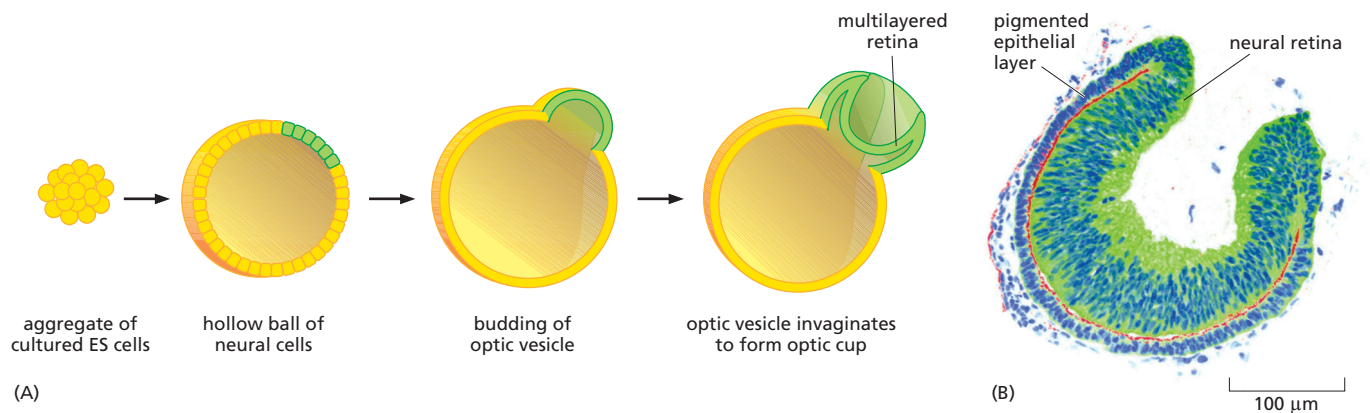


Figure 22–32 Cultured human ES cells can give rise to a three-dimensional organoid. (A) Schematic drawing shows how, under appropriate conditions, mouse or human pluripotent stem cells and their progeny in culture can proliferate, differentiate, and self-assemble into a three-dimensional eye-like structure (an optic cup), which includes a multilayered retina similar in organization to the one that forms during normal eye development *in vivo*. (B) Fluorescence micrograph of an optic cup formed by human ES cells in culture. The structure includes a developing retina containing multiple layers of neural cells (stained green) and an underlying layer of pigmented epithelium, the apical surface of which is stained red. All nuclei are stained blue. (A, adapted from M. Eiraku and Y. Sasai, *Curr. Opin. Neurobiol.* 22:768–777, 2012; B, from T. Nakano et al., *Cell Stem Cell* 10:771–785, 2012. With permission from Elsevier.)

regulators—not Oct4, Sox2, Klf4, and Myc, but Gata4, Mef2c, and Tbx5—it is possible to convert heart fibroblasts directly into heart muscle cells. This has been done in the living mouse, using retroviral vectors, and the transformation occurs with high efficiency when the vectors carrying the transgenes are injected directly into the heart muscle tissue itself. Although fibroblasts occupy only a small fraction of the heart tissue volume, they outnumber the heart muscle cells in the normal heart, and they survive in large numbers in areas of the heart where heart muscle cells have died. Thus, in a typical nonfatal heart attack, when heart muscle cells have died for lack of oxygen, the fibroblasts proliferate and make collagenous extracellular matrix, replacing the lost muscle with a fibrous scar. This is a poor sort of repair. By forcing expression of the appropriate factors in the heart, as described above, it has proved possible, in the mouse at least, to do better than nature and regenerate lost heart muscle through transdifferentiation of heart fibroblasts.

We are still a long way from putting this technique into practice as a treatment for heart attacks in humans, but it shows what the future may hold—not only for this medical problem but also for many others.

ES and iPS Cells Are Also Useful for Drug Discovery and Analysis of Disease

A large part of the excitement surrounding ES and iPS cells and the technology of transdifferentiation comes from the prospect of using the artificially generated cells for tissue repair. It begins to seem that virtually any type of tissue might be replaceable, allowing treatment of degenerative diseases that have previously had no cure, other than by organ transplantation. Research in this area is moving rapidly, but there are many difficulties to be overcome.

With the advent of iPS cells and direct transdifferentiation, at least one major hurdle that has challenged organ transplantation has been surmounted, in principle at least: the problem of immune rejection. ES cells, because they are created from early embryos that generally come from unrelated donors, will never be genetically identical to the cells of the individual receiving the cell transplant. The transplanted cells and their progeny are therefore prone to rejection by the immune system. Both iPS and transdifferentiated cells, in contrast, can be generated from a small sample of the individual's own tissue and so should escape immune attack when transplanted back into the same individual.

Tissue repair by cell and organoid transplantation, however, is not the only application for which ES, iPS, and transdifferentiated cells can be used: there are other ways that promise to be more immediately valuable. In particular, the cells can be used to generate large, homogeneous populations of specialized cells of any chosen type in culture, and these can serve both for investigation of disease mechanisms and for the search for new drugs acting on a specific cell defect (Figure 22–33).

Where a disease has a genetic cause, one can derive iPS cells from an affected individual and use the cells to produce the specific cell types that malfunction, to investigate how the malfunction occurs, and to screen for drugs that might help to put it right. *Timothy syndrome* provides an example. In this rare genetic condition, there is a severe, life-threatening disorder in the rhythm of the heartbeat (as well as several other abnormalities) as a result of a mutation in a specific type of Ca^{2+} channel. To study the underlying pathology, researchers took skin fibroblasts from individuals with the disorder, generated iPS cells from the fibroblasts, and induced the iPS cells to differentiate into heart muscle cells. These cells, when compared with heart muscle cells prepared similarly from normal control individuals, showed irregular heart muscle contractions and abnormal patterns of Ca^{2+} influx and electrical activity that could be characterized in detail. From these findings, it is a small step to development of an *in vitro* assay for drugs that might correct the misbehavior of the heart muscle cells.

Even in human disorders without a known genetic cause, iPS cells can be useful in understanding the condition. In one example, brain organoids were

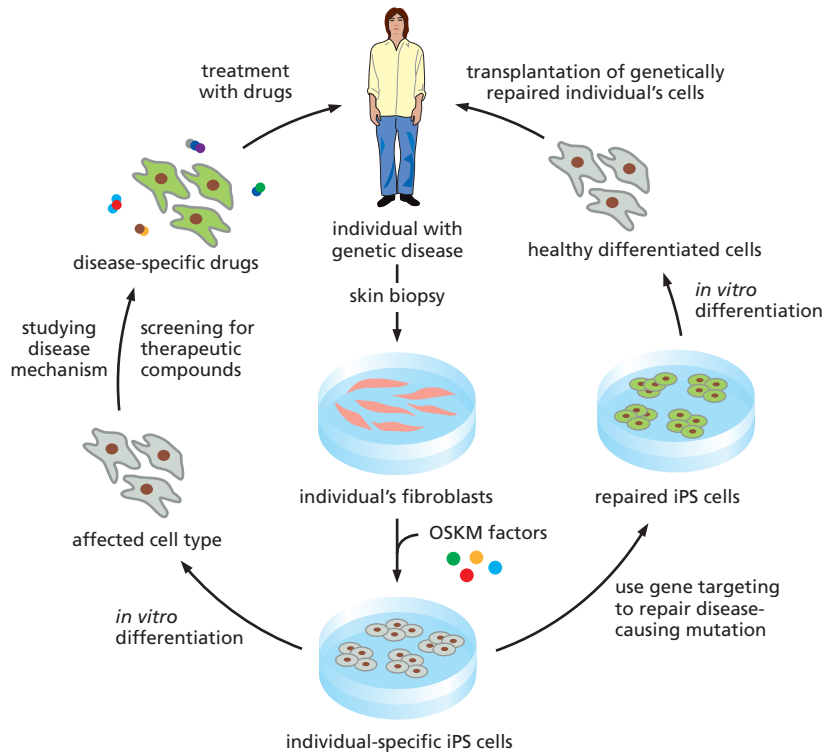


Figure 22–33 Use of iPS cells for drug discovery and for analysis and treatment of genetic disease. The left side of the diagram shows how differentiated cells that are generated from iPS cells derived from an individual with a genetic disease can be used for analysis of the disease mechanism and for discovery of therapeutic drugs. The right side of the diagram shows how the genetic defect might be repaired in the iPS cells, which could then be induced to differentiate in an appropriate way and grafted back into the individual without danger of immune rejection. (Adapted from D.A. Robinton and G.Q. Daley, *Nature* 481:295–305. Reproduced with permission of SNCSC.)

produced using iPS cells from an individual with microcephaly, a condition characterized by severely stunted brain growth and development. Careful analysis of these developing brain organoids revealed that the microcephaly in this case seemed to be caused by the premature cessation of proliferation and differentiation of brain progenitor cells, resulting in abnormally small numbers of differentiated brain cells.

Summary

In the adult mammalian body, the various types of tissue-specific stem cells are highly specialized, each giving rise to a limited range of differentiated cell types. Cells become restricted to specific pathways of differentiation during embryonic development. One way to force a return to a pluripotent or totipotent state is by nuclear transplantation: the nucleus of a differentiated cell can be injected into an enucleated oocyte, whose cytoplasm reprograms the genome of the injected nucleus back to an approximation of an early embryonic state. This allows the injected oocyte to develop into an entire new individual. The reversion of the genome to this state involves radical, genome-wide changes in chromatin structure and DNA methylation.

Remarkably, cells taken from the inner cell mass of an early mammalian embryo can be propagated in culture indefinitely in a pluripotent state. When transplanted back into a host early embryo, these embryonic stem (ES) cells can contribute cells to any tissue, including the germ line. ES cells have been invaluable for genetic engineering in mice. Cells with similar properties, called induced pluripotent stem (iPS) cells, can be generated from adult differentiated cells such as fibroblasts by forced expression of a cocktail of key transcription regulators. A similar method can be used to reprogram differentiated adult cells directly from one specialized cell type to another. In principle, iPS cells generated from cells taken from an adult human individual could be used for tissue repair in that same individual, avoiding the problem of immune rejection. More immediately, iPS cells provide a source of specialized cells that can be used to analyze in vitro the effects of mutations affecting human cells and to screen for drugs for treatment of genetic diseases. Both ES and iPS cells and their progeny can form tiny organs (organoids) in culture, which can serve as powerful models for studies of human development and disease.

PROBLEMS

Which statements are true? Explain why or why not.

22-1 In the mouse small intestine, stem cells in the crypts divide asymmetrically to maintain the population of cells that make up the crypts and villi; after each division, one daughter remains a stem cell and the other begins to divide rapidly to produce differentiated progeny.

22-2 Stem cells are the same in all tissues.

22-3 Every tissue that can be renewed is renewed from a tissue-specific population of stem cells.

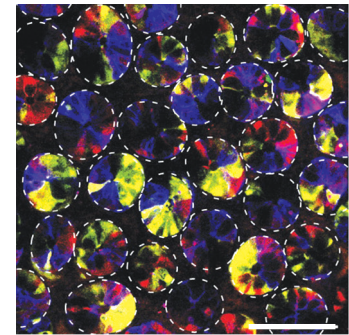
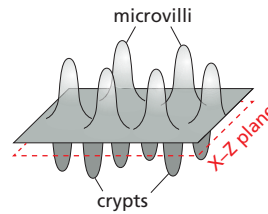
22-4 Although mice continually replace neurons in their olfactory bulbs and songbirds replace large numbers of neurons to refine their songs for each mating season, humans do not have the ability to replace neurons in their brains.

Discuss the following problems.

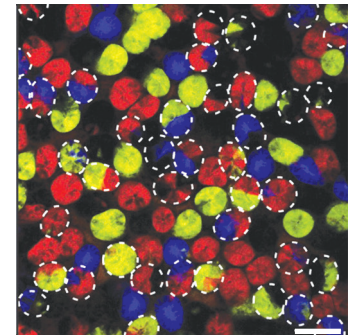
22-5 In the 1950s, scientists fed ^3H -thymidine to rats to label cells that were synthesizing DNA, and then followed the fates of labeled cells for periods of up to a year. They found three patterns of cell labeling in different tissues. Cells in some tissues such as neurons in the central nervous system and the retina did not get labeled. Muscle, kidney, and liver, by contrast, each showed a small number of labeled cells that retained their label, apparently without further division or loss. Finally, cells such as those in the squamous epithelia of the tongue and esophagus were labeled in fairly large numbers, with radioactive pairs of nuclei visible in 12 hours; however, the labeled cells disappeared over time. Which of these three patterns of labeling would you expect to see if the labeled cells were generated by stem cells? Explain your answer.

22-6 At any given time, a single intestinal crypt of mice comprises about 15 stem cells and 10 Paneth cells. After cell division, which occurs about once a day, the daughter cells remain stem cells only if they maintain contact with a Paneth cell. This constant competition for Paneth-cell contact raises the possibility that crypts might become monoclonal over time; that is, the crypt cells at one point in time might derive from only one of the 15 stem cells that existed at some earlier time. To test this possibility, you use the so-called confetti marker that upon activation expresses one of three fluorescent proteins in the stem cells of the crypt. You then examine crypts at various times to determine whether they contain cells with multiple colors or only one color (**Figure Q22-1**). Do the crypts become monoclonal over time or not? How can you tell?

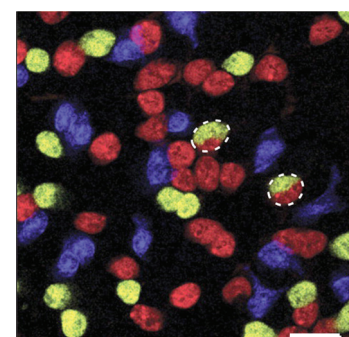
22-7 The origin of new β cells of the pancreas—from stem cells or from preexisting β cells—was not resolved until recently, when the technique of lineage tracing was used to decide the issue. Transgenic mice were engineered to express a tamoxifen-activated form of Cre recombinase under the control of the insulin promoter, which is



4 days



4 weeks



30 weeks

Figure Q22-1 Fluorescent cells in crypts in mouse intestines at various times after activation of expression of fluorescent proteins (Problem 22-6). The images are taken in the X-Z plane, which cuts through multiple crypts, as indicated in the schematic drawing. Roughly 50 crypts are visible in each section. Dotted white circles identify some individual crypts. Scale bars are 100 μm . (Adapted from H.J. Snippert et al., *Cell* 143:134–144, 2010. With permission from Elsevier.)

active only in β cells. In these mice, investigators could remove an inhibitory segment of DNA by adding tamoxifen and thereby allow expression of human placental alkaline phosphatase (HPAP), which can be detected by histochemical staining. After a pulse of tamoxifen that converted about 30% of β cells in young mice to cells that express HPAP, the investigators followed the percentage of labeled β cells for a year, during which time the total number of β cells in the pancreas increased by 6.5-fold. How do you suppose the percentage of β cells would change over time if new β cells were derived from stem cells? What if new β cells were derived from preexisting β cells? Which hypothesis do the results in **Figure Q22-2** support?

22-8 One of the earliest assays for hematopoietic stem cells made use of their ability to form colonies in the spleens of heavily irradiated mice. By varying the amounts of transplanted bone marrow cells, investigators showed that the number of spleen colonies varied linearly with dose and that the curve passed through the origin,

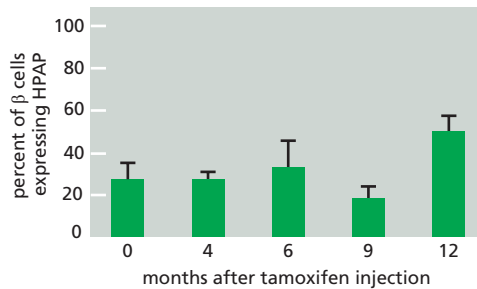


Figure Q22-2 Percentage of labeled β cells in pancreatic islets of mice at different ages (Problem 22-7). All mice were injected with a pulse of tamoxifen at 6–8 weeks of age, and then their pancreatic cells were stained for HPAP at various times afterward. Error bars represent standard deviations for all animals analyzed for that time point. (Adapted from Figure 2 of Y. Dor et al., *Nature* 429:41–46, 2004.)

suggesting that single cells were capable of forming individual colonies. However, because colony formation was rare relative to the numbers of transplanted cells, it was possible that undispersed clumps of two or more cells were the actual initiators.

A classic paper resolved this issue by exploiting rare, cytologically visible genome rearrangements generated by irradiation. Recipient mice were first irradiated to deplete bone marrow cells, and then they were irradiated a second time after transplantation to generate rare genome rearrangements in the transplanted cell population. Spleen colonies were then screened to find ones that carried genome rearrangements. How do you suppose this experiment distinguishes between colonization by single cells versus cellular aggregates?

22-9 It is possible to purify hematopoietic stem cells using a combination of antibodies directed against cell-surface targets. By removing cells that expressed surface markers characteristic of specific lineages such as B cells, granulocytes, myelomonocytic cells, and T cells, investigators generated a population of cells enriched for stem cells. They further enriched this population for putative stem cells by positively selecting for cells that expressed suspected stem-cell surface markers. Spleen colony formation in irradiated mice by these putative stem cells and the unfractionated bone marrow cells is shown in **Figure Q22-3**. Given that only about 1 in 10 cells lodges in the spleen, do these results support the idea that the enriched population consists mostly of hematopoietic stem cells? What additional information would you need to have to feel confident that the enriched cells are true stem cells? What proportion of bone marrow cells are hematopoietic stem cells?

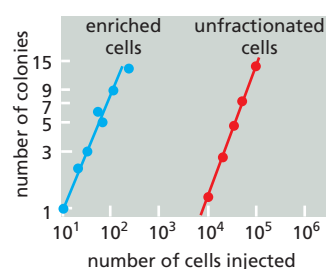


Figure Q22-3 Spleen colony formation by cells enriched for stem cells and by unfractionated bone marrow cells (Problem 22-9).

22-10 Generation of induced pluripotent stem (iPS) cells was first accomplished using retroviral vectors to carry the OSKM (Oct4, Sox2, Klf4, and Myc) set of transcription regulators into cells. The efficiency of fibroblast reprogramming was typically low (0.01%), in part because large numbers of retroviruses must integrate to bring about reprogramming, and each integration event carries with it the risk of inappropriately disrupting or activating a critical gene. In what other ways, or other forms, do you suppose you might deliver the OSKM transcription regulators so as to avoid these problems?

22-11 To test whether blood-borne factors alter neurogenesis in the central nervous system in mice, you use the technique of parabiosis—linking of circulatory systems in two individuals—to measure effects on the neurogenic niche in the dentate gyrus of the hippocampus. As indicated in **Figure Q22-4A**, you link the circulatory systems of two young mice, or two old mice, or a young mouse with an old mouse. Five weeks later, you stain slices of the dentate gyrus with antibodies to doublecortin (Dcx), a marker for newly born neurons, and count the new neurons, as summarized in **Figure Q22-4B and C**. Results in young-young parabionts and old-old parabionts were no different than in individual young or old mice. Do these results support the idea that blood-borne factors affect neurogenesis in the dentate gyrus of the hippocampus? Why or why not? Which, if either, mouse benefits in the parabiosis of a young mouse with an old mouse?

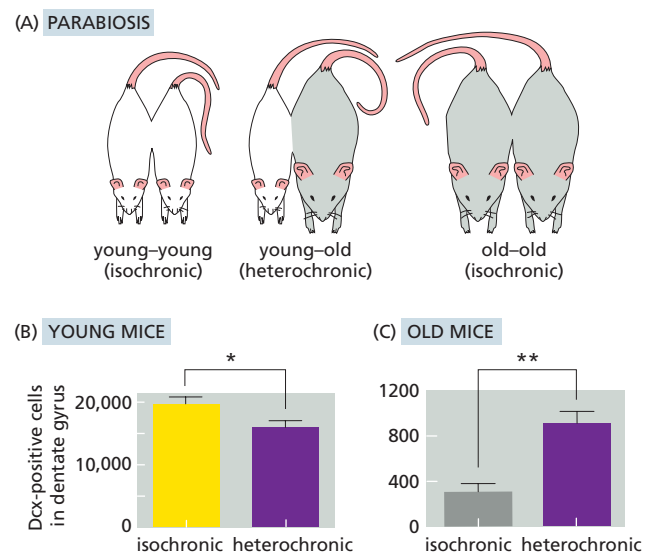


Figure Q22-4 Effects of parabiosis on the number of newly born neurons in the dentate gyrus of the mouse hippocampus (Problem 22-11).

(A) Linked circulatory systems (parabiosis) between pairs of mice. (B) Dcx-positive cells (newly born neurons) in a young mouse paired with a young mouse (isochronic) or in a young mouse paired with an old mouse (heterochronic). (C) Dcx-positive cells in an old mouse paired with an old mouse (isochronic) or in an old mouse paired with a young mouse (heterochronic). Isochronic refers to parabiosis between mice of the same age; heterochronic refers to parabiosis between mice of different ages. Asterisks refer to statistical significance of the results: * is for $P < 0.05$, which means the result would be expected to occur by chance in less than 1 in 20 repeats; ** is for $P < 0.01$, which means the result would be expected to occur by chance in less than 1 in 100 repeats.

REFERENCES

General

- Gurdon JB & Melton DA (2008) Nuclear reprogramming in cells. *Science* 322, 1811–1815.
- He S, Nakada D & Morrison SJ (2009) Mechanisms of stem-cell renewal. *Annu. Rev. Cell Dev. Biol.* 25, 377–406.
- Karagiannis P, Takahashi K, Saito M . . . Osafune K (2019) Induced pluripotent stem cells and their use in human models of disease and development. *Physiol. Rev.* 99, 79–114.
- Morrison SJ & Spradling AC (2008) Stem cells and niches: mechanisms that promote stem cell maintenance throughout life. *Cell* 132, 598–611.

Stem Cells and Tissue Homeostasis

- Barker N, van Es JH, Kuipers J . . . Clevers H (2007) Identification of stem cells in small intestine and colon by marker gene *Lgr5*. *Nature* 449, 1003–1007.
- Baron CS & van Oudenaarden A (2019) Unravelling cellular relationships during development and regeneration using genetic lineage tracing. *Nat. Rev. Mol. Cell Biol.* 20, 753–765.
- Blanpain C & Fuchs E (2014) Plasticity of epithelial stem cells in tissue regeneration. *Science* 344, 1242281.
- Morrison SJ, Uchida N & Weissman IL (1995) The biology of hematopoietic stem cells. *Annu. Rev. Cell Dev. Biol.* 11, 35–70.
- Taub R (2004) Liver regeneration: from myth to mechanism. *Nat. Rev. Mol. Cell Biol.* 5, 836–847.

Control of Stem-Cell Fate and Self-Renewal

- Crane GM, Jeffery E & Morrison SJ (2017) Adult haematopoietic stem cell niches. *Nat. Rev. Immunol.* 17(9), 573–590.
- Goodell MA & Rando TA (2015) Stem cells and healthy aging. *Science* 350, 1199–1204.
- Losick VP, Morris LX, Fox DT & Spradling A (2011) *Drosophila* stem cell niches: a decade of discovery suggests a unified view of stem cell regulation. *Dev. Cell* 21, 159–171.
- Sato T, Van Es JH, Snippert HJ . . . Clevers H (2011) Paneth cells constitute the niche for *Lgr5* stem cells in intestinal crypts. *Nature* 469, 415–418.
- Sunchu B & Cabernard C (2020) Principles and mechanism of asymmetric cell division. *Development* 147(13), dev167650.

- Venkei ZG & Yamashita Y (2018) Emerging mechanisms of asymmetric stem cell division. *J. Cell Biol.* 217(11), 3785–3795.

- Yoshida S (2019) Heterogeneous, dynamic, and stochastic nature of mammalian spermatogenic stem cells. *Curr. Top. Dev. Biol.* 135, 245–285.

Regeneration and Repair

- Brookes JP & Kumar A (2008) Comparative aspects of animal regeneration. *Annu. Rev. Cell Dev. Biol.* 24, 525–549.
- Tanaka EM (2016) The molecular and cellular choreography of appendage regeneration. *Cell* 165(7), 1598–1608.
- Tanaka EM & Reddien PW (2011) The cellular basis for animal regeneration. *Dev. Cell* 21, 172–185.
- Wagner DE, Wang IE & Reddien PW (2011) Clonogenic neoblasts are pluripotent adult stem cells that underlie planarian regeneration. *Science* 332, 811–816.

Cell Reprogramming and Pluripotent Stem Cells

- Apostolou E & Hochedlinger K (2013) Chromatin dynamics during cellular reprogramming. *Nature* 502, 462–471.
- Fox IJ, Daley GQ, Goldman SA . . . Trucco M (2014) Stem cell therapy. Use of differentiated pluripotent stem cells as replacement therapy for treating disease. *Science* 345, 1247391.
- Hoechedlinger K & Jaenisch R (2015) Induced pluripotency and epigenetic reprogramming. *Cold Spring Harb. Perspect. Biol.* 7, a019448.
- Inoue H, Nagata N, Kurokawa H & Yamanaka S (2014) IPS cells: a game changer for future medicine. *EMBO J.* 33, 409–417.
- Radziszewska A & Silver JCR (2014) Do all roads lead to Oct4? The emerging concepts of induced pluripotency. *Trends Cell Biol.* 24, 275–284.
- Sasai Y, Eiraku M & Suga H (2012) *In vitro* organogenesis in three dimensions: self-organising stem cells. *Development* 139, 4111–4121.
- Takahashi K & Yamanaka S (2006) Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors. *Cell* 126, 663–676.
- Theunissen TW & Jaenisch R (2014) Molecular control of induced pluripotency. *Cell Stem Cell* 14, 720–734.